

PLATFORMS / ASSAYS / TRANSLATION / PARTNERING

Biotechnology English

Explain the signal. Define the claim.
Connect evidence to decisions.

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Eight lessons. Eleven extended conversations.

INSIDE THE BOOK

192 vocabulary entries / 128 reusable phrases
220 extended-dialogue turns / 11 transfer exchanges
Word banks and explained answer keys

Upper-intermediate to advanced (B2-C1). For biotechnology research, translational science, development, operations, and partnering teams.

2026 edition

Eight scenarios. Eight skills.

Eight conversations connecting laboratory evidence, development decisions, and collaboration terms.

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A promising signal needs a precise claim.

Biotechnology teams must explain specialized results to colleagues who make different kinds of decisions. A scientist asks whether the assay supports the mechanism; a development lead asks what remains untested; a partner asks what rights and evidence are actually available. The conversations here keep the scientific detail while making those decision boundaries explicit.

Name the experimental system

State which model, material, readout, and comparison support the finding. A result in one setting can justify further investigation without proving a platform works across every proposed use.

"We observed the signal in one cell line; the broader application remains a hypothesis."

Expand the word validated

Say what was evaluated, for which intended use, against which criteria, and with what remaining limits. Analytical performance and a useful clinical decision are different claims.

"The assay measures the analyte in the assessed range; clinical prediction has not been established."

Make the next decision visible

A list of completed experiments is not a decision-ready update. Connect each result to the milestone criterion, the unresolved question, and the proposed next allocation of time or resources.

"The binding work is complete; the functional criterion remains open."

Define rights before promising access

Materials, confidential information, patents, know-how, and resulting data are different things. Identify the proposed use and unresolved terms before assuming that one agreement grants every permission.

"The confidentiality agreement does not settle the material-use or publication terms."

PROFESSIONAL BOUNDARIES

Original fictional language practice, not experimental instructions or medical, regulatory, legal, or investment advice. All organizations, candidates, numerical examples, and negotiations are invented. Apply current scientific standards, local procedures, appropriate professional review, and the actual governing agreements.

One cell line, three proposed diseases

A Read the situation

Distinguish a measured platform signal from a proposed mechanism and a broader development hypothesis.

Research lead Nia reviews a fictional platform presentation with venture lead Leo. Three independent preparations of one cell line show an average reporter-signal reduction of 40% against the stated control. No orthogonal functional result or data from other disease systems are available. The slide claims proven activity across three diseases. The immediate request is funding for a bounded follow-up study, not approval for clinical use. Nia and Leo must retain the promising observation while correcting the unsupported breadth.

PEOPLE / Nia: Research lead | Leo: Venture lead

PRECISION CAN STRENGTHEN A PITCH

A confident scientific presentation distinguishes observation, interpretation, and the next test. Qualifying a claim does not require burying the result. Put the system and limitation beside the headline so a listener can understand both the promise and the reason for the next study.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which evidence is supplied?

- | | |
|---|---|
| A Confirmed benefit in three diseases | B A completed human efficacy study |
| C Functional confirmation from an orthogonal assay | D A 40% average reporter-signal reduction in three independent preparations of one cell line |

2. Which claim exceeds that evidence?

- | | |
|--|--|
| A A signal observed in the stated cell line | B Three independent preparations tested |
| C A follow-up study proposed | D Proven activity across three diseases |

3. What is the immediate decision request?

- | | |
|--|---|
| A Authorizing clinical use | B Declaring every platform application validated |
| C Licensing all disease rights without review | D Funding a bounded follow-up study |

Answers and explanations: [page 85](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

platform technology

A reusable technological approach intended to support multiple applications.

IN USE develop a platform technology

scientific thesis

The evidence-based argument and hypotheses underlying a research program.

IN USE articulate the scientific thesis

modality

A type of therapeutic or technological approach.

IN USE compare therapeutic modalities

target

The biological entity a program aims to affect.

IN USE define the target

target engagement

Evidence that an intervention interacts with its intended biological target.

IN USE assess target engagement

mechanism of action

The biological process through which an intervention produces an effect.

IN USE investigate the mechanism of action

reporter signal

A measurable output used as an indicator in an experimental system.

IN USE quantify a reporter signal

cell line

A maintained population of cells used for research.

IN USE characterize the cell line

biological replicate

An independently prepared biological sample or experimental unit.

IN USE include biological replicates

technical replicate

A repeated measurement of the same biological material or preparation.

IN USE distinguish technical replicates

control condition

A comparison used to interpret an experimental result.

IN USE specify the control condition

orthogonal assay

A different measurement approach used to examine the same underlying question.

IN USE run an orthogonal assay

Precision check

Three independent preparations provide information about variation within the tested biological system. They are not three diseases or three clinical trials. A reporter signal can support a research direction without establishing the full mechanism or proving human benefit.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

functional readout

A measurement of a relevant biological activity or outcome.

IN USE confirm a functional readout

assay interference

An effect on the measurement system that can distort the apparent result.

IN USE investigate assay interference

counter-screen

A separate screen used to identify unwanted or misleading effects.

IN USE include a counter-screen

selectivity

Preference for the intended target or effect relative to alternatives.

IN USE assess selectivity

off-target effect

An effect involving an unintended biological target or pathway.

IN USE evaluate off-target effects

proof of concept

Evidence supporting feasibility within a specified scope.

IN USE define the proof-of-concept scope

proof of mechanism

Evidence supporting the proposed causal biological mechanism.

IN USE seek proof of mechanism

generalizability

The extent to which a finding applies beyond the tested setting.

IN USE assess generalizability

disease model

An experimental system representing selected aspects of a disease.

IN USE qualify the disease model

translational hypothesis

A proposed link between experimental findings and a later development setting.

IN USE test a translational hypothesis

evidence gap

A missing piece of information relevant to a claim or decision.

IN USE identify an evidence gap

decision gate

A defined point at which evidence is reviewed against criteria.

IN USE set a decision gate

Precision check

An orthogonal assay uses a different measurement approach to examine the underlying question; a technical repeat of the same readout is not automatically orthogonal. Define the relevant follow-up and decision criteria without assuming its result in advance.

Phrases you can actually use

C Notice the language / 1 of 2

LEAD WITH THE OBSERVATION

"We observed a 40% average reduction in the reporter signal."

LIMIT THE BREADTH

"Other disease systems have not yet been tested."

KEEP THE HYPOTHESIS

"The broader platform application remains a testable hypothesis."

CHECK INTERFERENCE

"Could the signal reflect an effect on the assay rather than the intended biology?"

NAME THE SYSTEM

"The data come from three independent preparations of one cell line."

SEPARATE SIGNAL FROM FUNCTION

"The reporter result does not yet establish the functional effect."

REQUEST ANOTHER METHOD

"We need a different readout to examine the underlying question."

DESCRIBE THE CONTROL

"The percentage is relative to the control specified in this experiment."

Promising

Signals potential without claiming an established development outcome.

Proven

Requires the claim's scope to match the supporting evidence.

Platform

A reusable ambition does not itself demonstrate every application.

Phrases you can actually use

C Notice the language / 2 of 2

AVOID A MECHANISM CLAIM

"We have not established the full mechanism from this result alone."

IDENTIFY THE GAP

"The missing evidence is functional confirmation in a relevant additional system."

KEEP REPLICATION PRECISE

"These are independent preparations, not repeated reads presented as independent samples."

BOUND THE ASK

"We are requesting the next research step, not clinical authorization."

STATE THE OPPORTUNITY

"The signal is promising enough to justify a focused next study."

DEFINE THE DECISION

"The next study should answer whether this application merits further development."

CORRECT THE SLIDE

"Replace proven across three diseases with the result actually measured."

CLOSE WITH CRITERIA

"Agree what evidence would support or stop the next stage."

Replicate

Specify whether the repeat is technical, biological, or across runs.

Reduction

Name the readout and comparison; do not let a percentage imply patient benefit.

Orthogonal

Different measurement logic matters more than a new label on the same assay.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which slide headline is best supported?

- A** All off-target effects excluded.
- B** Reporter signal reduced by an average 40% in the tested cell line; broader applications remain untested.
- C** Platform efficacy proven across three diseases.
- D** Human benefit confirmed by three preparations.

2. One preparation is read three times with the same method. How should those reads be described?

- A** Three biological replicates from three separate preparations
- B** Three orthogonal assays confirming the mechanism
- C** Three disease models demonstrating broader applicability
- D** Three technical measurements of one biological preparation

3. Why request an orthogonal functional readout?

- A** To guarantee a favorable outcome
- B** To avoid describing the original control
- C** To examine the biological question through a different measurement approach
- D** To relabel the same reporter result as clinical evidence

4. Which funding request matches the brief?

- A** Authorize clinical use because the signal is promising.
- B** Assume all diseases will respond and skip further evidence.
- C** Treat three preparations as completed trials in three indications.
- D** Support a bounded follow-up addressing functional confirmation and broader relevance.

[Answers and explanations: page 85.](#)

One cell line, three proposed diseases

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

scientific thesis / biological replicate / evidence gap / orthogonal assay / generalizability / proof of mechanism / reporter signal / translational hypothesis / assay interference / decision gate

- 01 **Leo:** The headline says our platform has proven activity across three diseases. It sounds strong, but I want to know whether each part of that claim is supported.
- 02 **Nia:** We have a ¹ _____ reduction averaging forty percent against the specified control. The work used three independent preparations of one cell line, not three disease systems.
- 03 **Leo:** Can we still call it a platform result? The investors need to see why this is more than a single isolated observation with no development direction.
- 04 **Nia:** Yes, as evidence relevant to the ² _____. We should explain the proposed applications while separating what we observed from what we expect to test next.
- 05 **Leo:** The slide currently turns the three preparations into a visual with three disease labels. I can see how that could imply that each disease has been tested.
- 06 **Nia:** Keep the ³ _____ description explicit. Three independent preparations tell us something about that system; they do not create evidence for three separate indications.
- 07 **Leo:** What would you say about the mechanism? The lower signal looks consistent with the proposed pathway, but the headline makes the connection sound settled.
- 08 **Nia:** We do not yet have ⁴ _____. The reporter finding supports investigation, but an effect on the measurement system could also influence the apparent result.
- 09 **Leo:** What would the next study tell us that another reporter run would not? The funding request needs a scientific question, not just a larger sample count.
- 10 **Nia:** It should include an ⁵ _____ addressing the underlying biology. A different measurement approach would help distinguish a relevant biological effect from a feature of this reporter system.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 10.

- 11 **Leo:** We should be clear that you are asking to test that distinction, not promising that the second method will confirm the first. Otherwise the milestone is misleading.
- 12 **Nia:** Correct. We also need to assess ⁶ _____ and the appropriate controls. A promising decrease is not sufficient to dismiss competing explanations for the measured change.
- 13 **Leo:** How do we describe the proposed disease expansion without making it sound like a result? I want an ambitious but defensible sentence in the investor discussion.
- 14 **Nia:** Call it a ⁷ _____ and name the additional systems we propose to evaluate. The audience can understand the opportunity without being told that unperformed work has succeeded.
- 15 **Leo:** I will keep the forty-percent observation in the main line and put the tested system beside it. The expansion will be a proposed application, not a proven one.
- 16 **Nia:** That makes the ⁸ _____ visible. We still lack functional confirmation and data in other disease systems, so those limits should travel with the headline.
- 17 **Leo:** For the decision itself, we want a bounded research budget, not a commitment to take every suggested indication into development. The proposal needs a stopping point.
- 18 **Nia:** Set a ⁹ _____ with agreed evidence criteria. We should know what question the work will answer and how that answer affects the next development choice.
- 19 **Leo:** I will remove the three-disease claim and request the bounded follow-up. Keep the forty-percent result beside the actual cell system so the claim cannot drift again.
- 20 **Nia:** Good. That preserves the platform ambition while keeping ¹⁰ _____ open to investigation. We can be confident about the observation without claiming that every future application already works.

ANSWERS / EXPLANATIONS: PAGE 86

1. reporter signal | 2. scientific thesis | 3. biological replicate | 4. proof of mechanism | 5. orthogonal assay | 6. assay interference | 7. translational hypothesis | 8. evidence gap | 9. decision gate | 10. generalizability

One model is not every model

F Say it

A screening result is observed in one cell model. Two other proposed models have not been tested. The team requests a study of those additional systems.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

model / hypothesis / gap / untested

Scientist: "The signal was observed in one ¹ _____."

Partner: "The other two are still ² _____."

Scientist: "Broader application remains a ³ _____."

Partner: "The next study should address that evidence ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 86

1. model | 2. untested | 3. hypothesis | 4. gap

Rehearse with a fixed script

1. Read the slide-review exchange. State 40% average reporter reduction, three independent preparations, and one cell line together.
2. Swap roles. Contrast a technical repeat with an orthogonal method; retain the untested broader applications.
3. Read the corrected one-model transfer. Keep two proposed models untested and the wider claim a hypothesis.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Tight wells, shifting days

A Read the situation

Explain within-run precision separately from between-day variation and request a traceable investigation.

Assay scientist Imani and project lead Chen review a fictional control sample measured over three days. Each day has a within-run coefficient of variation of 5%, but the day means are 80, 100, and 125 signal units. The same nominal control level was intended each day. The summary says fully reproducible. Reagent-lot, calibration, and operator records still need comparison. No acceptance limits or confirmed cause are supplied; the team must describe the variation without inventing a validation decision.

PEOPLE / Imani: Assay scientist | Chen: Project lead

ASK WHICH KIND OF CONSISTENCY

A colleague may use reproducible as a broad compliment while an assay specialist means a defined set of conditions. Clarify the tested source of variation rather than argue about the word alone. Numbers become useful when the comparison, units, and experimental structure are visible.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is consistent within each run?

- A** A common acceptance limit has been met.
- B** The stated coefficient of variation is 5%.
- C** Every day mean is 100.
- D** Every reagent lot has been verified identical.

2. What changes across days?

- A** The mean signal: 80, 100, and 125 units
- B** The supplied acceptance limit
- C** The number of confirmed disease indications
- D** The definition of an independent clinical trial

3. What cannot yet be concluded?

- A** A confirmed cause or full validation status
- B** That day three has a mean of 125
- C** That the within-run CV is supplied
- D** That records need comparison

Answers and explanations: page 87.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

assay development

Designing and refining a measurement method for a specified purpose.

IN USE plan assay development

repeatability

Precision under closely matched conditions over a short interval.

IN USE assess repeatability

intermediate precision

Precision with relevant changes within one laboratory, such as day or operator.

IN USE evaluate intermediate precision

reproducibility

Precision under specified changed conditions, commonly across laboratories.

IN USE define reproducibility conditions

within-run variation

Differences among measurements within a single analytical run.

IN USE quantify within-run variation

between-day variation

Differences associated with measurements on different days.

IN USE investigate between-day variation

coefficient of variation

Standard deviation divided by the mean, often expressed as a percentage.

IN USE report the coefficient of variation

standard deviation

A measure of how observations are spread around their mean.

IN USE calculate the standard deviation

mean signal

The average measured output for a defined group.

IN USE compare mean signals

precision

Closeness among repeated measurements under stated conditions.

IN USE assess analytical precision

accuracy

Closeness of a measurement to the relevant reference or accepted value.

IN USE evaluate analytical accuracy

bias

A systematic difference from the relevant reference value.

IN USE investigate measurement bias

Precision check

A 5% within-run coefficient of variation describes relative spread inside that run. It does not establish matching means across days or accuracy against a reference. Without supplied acceptance criteria, the exercise cannot declare the assay validated or failed.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

calibration

Establishing the relationship between measurement response and reference values.

IN USE verify calibration records

reference standard

A characterized material used to support measurement comparisons.

IN USE qualify a reference standard

quality-control sample

A sample used to monitor analytical performance.

IN USE review quality-control samples

reagent lot

An identified production batch of a reagent.

IN USE compare reagent lots

lot-to-lot variation

Differences associated with separate batches of a material.

IN USE evaluate lot-to-lot variation

plate effect

Variation associated with a measurement plate or its positions.

IN USE assess a plate effect

edge effect

A position-related difference at the outer wells of a plate.

IN USE check for an edge effect

assay drift

A change in measurement performance over time.

IN USE monitor assay drift

normalization

Adjusting results to a defined comparison or reference basis.

IN USE document normalization

acceptance criterion

A predefined requirement for judging a result or process.

IN USE specify acceptance criteria

root-cause investigation

A structured effort to identify supported underlying explanations.

IN USE conduct a root-cause investigation

run metadata

Information describing the conditions and history of an analytical run.

IN USE preserve run metadata

Precision check

Intermediate precision examines relevant changes within a laboratory; reproducibility often refers to broader changed conditions. Specify the intended meaning. Normalization must have a justified, documented basis, not be invented after the fact solely to make disagreeing runs appear aligned.

Phrases you can actually use

C Notice the language / 1 of 2

SEPARATE THE LEVELS

"The within-run precision is consistent, but the day means differ."

DEFINE THE STATISTIC

"The CV describes relative spread within each run."

ASK FOR THE CRITERIA

"Which predefined acceptance criteria apply to this intended use?"

PRESERVE THE RAW RESULT

"Keep the original measurements and their run records."

GIVE THE VALUES

"The means are eighty, one hundred, and one hundred twenty-five signal units."

LIMIT THE CONCLUSION

"That does not establish agreement across days."

CHECK THE COMPARISON

"Was the same nominal control level used in each run?"

INVESTIGATE CONDITIONS

"Compare reagent lots, calibration, operator, and relevant run conditions."

Precise versus accurate

Close repeated values may still be systematically offset from a reference.

CV

Expand coefficient of variation before using the abbreviation in a mixed-audience meeting.

Same mean

Equal spread does not require equal averages.

Phrases you can actually use

C Notice the language / 2 of 2

AVOID PREMATURE BLAME

"We have not established which factor explains the shift."

NAME THE PERCENTAGE BASE

"Twenty-five above one hundred is a 25% increase relative to one hundred."

KEEP NORMALIZATION JUSTIFIED

"State the reference and method used for any adjustment."

REPORT THE STATUS

"Between-day variation is under investigation."

EXPLAIN PRECISION

"Tight measurements can still differ systematically from a reference."

CHALLENGE THE LABEL

"Fully reproducible is broader than these data support."

REQUEST TRACEABILITY

"Link the plotted mean to its raw data and metadata."

CLOSE WITH THE QUESTION

"The next review must explain the shift and evaluate it against the agreed criteria."

Drift

A descriptive possibility, not a confirmed mechanism for these three values.

Reproducible

Specify what was repeated and which conditions changed.

Failed

A formal judgment requires the applicable criteria, not only an unfavorable-looking chart.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which statement fits the data?

- A** The operator caused the shift, as already established.
- B** Each run has 5% CV, while day means differ from 80 to 125 units.
- C** A 5% CV proves the same mean every day.
- D** The assay is clinically validated because the means are positive.

2. Relative to day two's mean of 100, how much higher is day three's mean of 125?

- A** 25%
- B** 5%
- C** 20%
- D** 125%

3. Which follow-up best supports investigation?

- A** Compare traceable run conditions and raw data before assigning a cause.
- B** Delete the two means that differ from 100.
- C** Declare every run acceptable without criteria.
- D** Rename the axis to conceal the shift.

4. A colleague rescales each day to a mean of 100 and calls the assay reproducible. What remains unresolved?

- A** Whether the rescaled means are numerically equal to each other
- B** Whether the three original runs occurred on different days
- C** Whether the adjusted chart now uses the same vertical scale
- D** Whether the adjustment has a justified basis and the original between-day variation is understood

Answers and explanations: page 87.

Tight wells, shifting days

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

run metadata / calibration / within-run variation / root-cause investigation / coefficient of variation / intermediate precision / mean signal / acceptance criterion / normalization / precision

- 01 **Chen:** Every run has a five-percent CV, so I kept fully reproducible in the summary. What am I missing before we show this to the project team?
- 02 **Imani:** The ¹ _____ is similar, but the means are eighty, one hundred, and one hundred twenty-five. The runs do not sit at the same level.
- 03 **Chen:** So the points can be tightly grouped each day while the groups move? I treated the repeated percentage as evidence of an unchanged result.
- 04 **Imani:** The ² _____ is standard deviation divided by mean. Five percent describes relative spread; it does not tell us that the means match.
- 05 **Chen:** We intended the same nominal control level. Could the newest operator explain the shift? That was the first suggestion in yesterday's meeting.
- 06 **Imani:** Let us compare the ³ _____ first: operator, reagent lot, calibration and relevant conditions. We do not yet know which changes coincide.
- 07 **Chen:** Right. If the operator and reagent lot changed together, the pattern would not isolate either one. I should not report an attribution we cannot defend.
- 08 **Imani:** Keep both as questions in the ⁴ _____. The point is to test explanations against the records, not settle on the most convenient person to blame.
- 09 **Chen:** All three runs were in this laboratory. Is between-day reproducibility the clearest heading, or would another term make the conditions more precise?
- 10 **Imani:** We are examining ⁵ _____ within this laboratory. State the days and conditions examined so the reader does not infer an interlaboratory study.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 19.

- 11 **Chen:** I also wrote that day three rose twenty percent from day two. I divided twenty-five by the final value of one hundred twenty-five.
- 12 **Imani:** Use day two as the base. The ⁶ _____ increased twenty-five percent: twenty-five divided by one hundred. Twenty percent describes the reverse decrease.
- 13 **Chen:** Thanks. Is the shift enough to call the assay failed? I cannot find the intended-use limits in this review pack.
- 14 **Imani:** We need the applicable ⁷ _____. Do not invent a pass or failure threshold from this chart; report the between-day difference as under investigation.
- 15 **Chen:** Someone offered to rescale every day's mean to one hundred. It would remove the visible shift, but would it resolve the performance question?
- 16 **Imani:** Not by itself. Any ⁸ _____ needs a justified reference and documented method. An adjustment chosen only to align these means can hide what needs investigation.
- 17 **Chen:** Then retain the original readings and show how any adjusted result was derived. The unadjusted plot should not disappear from the package.
- 18 **Imani:** Agreed. Include the ⁹ _____ records too, so we can trace the measurement basis rather than infer it from the graph alone.
- 19 **Chen:** I will show the three means beside their CVs, correct the percentage, and list the missing criteria and records. No validation conclusion yet.
- 20 **Imani:** That gives ¹⁰ _____ its proper scope. We retain the useful within-run result while making the unresolved between-day performance visible.

ANSWERS / EXPLANATIONS: PAGE 88

1. within-run variation | 2. coefficient of variation | 3. run metadata | 4. root-cause investigation | 5. intermediate precision | 6. mean signal | 7. acceptance criterion | 8. normalization | 9. calibration | 10. precision

A percentage needs its reference

F Say it

Two comparable fictional runs have means of 120 and 150 units. Their within-run CVs are both 4%. No acceptance criteria are supplied.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

25% / spread / criteria / thirty

Analyst: "The mean rose by ¹ _____ units."

Reviewer: "Relative to 120, that is a ² _____ increase."

Analyst: "The four-percent CV describes within-run ³ _____."

Reviewer: "A formal acceptance decision still needs the applicable
⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 88

1. thirty | 2. 25% | 3. spread | 4. criteria

Rehearse with a fixed script

1. Read the assay review. State the 5% within-run CV alongside daily means of 80, 100, and 125.
2. Swap roles. Correct the denominator: 100 to 125 is a 25% increase; 125 to 100 is a 20% decrease.
3. Read the corrected transfer. Preserve the 120 reference, 30-unit change, 25% increase, and missing acceptance criteria.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Validated for which use?

A Read the situation

03

Specify the scope of biomarker validation and distinguish measurement performance from response prediction.

Translational lead Sara and assay lead Noah review a fictional plasma-protein test. The analytical study met its predefined measurement criteria over 2 to 20 research units in the tested plasma conditions. No study has established that the protein predicts response to candidate B17. A slide calls it a validated predictive biomarker and proposes selecting trial participants by its value. The team must correct that claim and identify the intended-use evidence still needed. No patient-selection rule is authorized by the exercise.

PEOPLE / Sara: Translational lead | Noah: Assay lead

A QUALIFIER SHOULD IDENTIFY THE CLAIM

Validated is persuasive because it sounds complete. Ask what was validated, in which material and range, and for which intended use. These questions help colleagues align scientific and clinical claims without treating the person who raised the issue as an obstacle to progress.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which claim is supported by the supplied study?

A Regulatory qualification has been granted.

B The test met predefined analytical criteria over the assessed range and plasma conditions.

C The protein predicts B17 response.

D A patient-selection threshold has clinical support.

2. What evidence is absent?

A The existence of predefined measurement criteria

B A study establishing prediction of response to B17

C The fact that plasma was tested

D The stated analytical range

3. Which proposed use requires additional support?

A Stating the tested range

B Reporting the analytical study's actual scope

C Selecting participants on an asserted predictive role

D Describing the measured analyte

Answers and explanations: [page 89](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

biomarker

A measured characteristic indicating a biological process or response.

IN USE evaluate a biomarker

analyte

The substance or component a test measures.

IN USE identify the analyte

context of use

The defined purpose and setting in which a biomarker is to be used.

IN USE specify the context of use

analytical validation

Establishing that a test adequately measures its intended analyte for the stated use.

IN USE complete analytical validation

clinical validation

Establishing the relationship to the relevant clinical concept for the proposed use.

IN USE plan clinical validation

clinical utility

The usefulness of applying a test in a particular care decision.

IN USE assess clinical utility

biomarker qualification

Formal acceptance for a specified use through an applicable qualification process.

IN USE define the qualification scope

fit for purpose

Adequate for a specifically defined intended use.

IN USE demonstrate fit-for-purpose performance

reportable range

The interval over which results can be appropriately reported under the method.

IN USE specify the reportable range

limit of detection

The lowest level distinguishable from absence under defined conditions.

IN USE establish a limit of detection

limit of quantification

The lowest level measurable with specified quantitative performance.

IN USE establish a limit of quantification

matrix effect

An influence of other sample components on the measurement.

IN USE assess matrix effects

Precision check

Analytical validation concerns the measurement; clinical validation concerns the proposed clinical relationship. Clinical utility asks a further question about using the result in a decision. None of these labels should be inferred solely from another label or from an attractive plot.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

diagnostic sensitivity

The proportion of people with the target condition correctly identified as positive.

IN USE estimate diagnostic sensitivity

diagnostic specificity

The proportion without the target condition correctly identified as negative.

IN USE estimate diagnostic specificity

positive predictive value

The proportion of positive test results that correspond to the target condition.

IN USE interpret positive predictive value

negative predictive value

The proportion of negative results that correspond to absence of the target condition.

IN USE interpret negative predictive value

prevalence

The proportion of the relevant population with a condition at a stated time.

IN USE account for prevalence

decision threshold

A specified cutoff used in making a defined classification or decision.

IN USE justify a decision threshold

prognostic biomarker

A marker associated with likelihood of a clinical event or disease progression.

IN USE distinguish a prognostic biomarker

predictive biomarker

A marker identifying differing likelihood of an effect from a particular intervention.

IN USE evaluate a predictive biomarker

pharmacodynamic biomarker

A marker indicating biological response after exposure to an intervention.

IN USE measure a pharmacodynamic biomarker

patient stratification

Grouping patients by defined characteristics for analysis or an intended use.

IN USE justify patient stratification

prospective validation

Testing performance in a study planned before the relevant outcomes are observed.

IN USE design prospective validation

external validation

Evaluation in an independent setting or data source.

IN USE obtain external validation

Precision check

Predictive refers here to differing effects of a particular intervention; prognostic concerns clinical course. Diagnostic sensitivity is not the same as an assay's detection limit. Always define the population, sample conditions, range, and intended decision before interpreting performance claims.

Phrases you can actually use

C Notice the language / 1 of 2

03

ASK FOR THE SCOPE

"Validated for which measurement and intended use?"

NAME THE RANGE

"The assessed measurement range is two to twenty research units."

SEPARATE THE CLAIMS

"Analytical performance does not establish prediction of treatment response."

AVOID PREMATURE SELECTION

"We do not yet have evidence supporting that predictive selection rule."

STATE THE ANALYTE

"The test measures the specified protein in plasma."

PRESERVE THE CONDITIONS

"That evidence applies to the tested plasma conditions."

DEFINE THE PURPOSE

"The proposed use is participant selection for the B17 program."

USE THE CORRECT LABEL

"Describe an analytically evaluated test, not a clinically validated predictive biomarker."

Validated

Name the object and use rather than presenting validation as unlimited.

Predictive

In this context, prediction concerns effects of a particular intervention.

Qualified

A formal process and its defined scope cannot be inferred from an internal label.

Phrases you can actually use

C Notice the language / 2 of 2

CLARIFY QUALIFICATION

"A study result is not automatically formal biomarker qualification."

SEPARATE PROGNOSIS

"Association with disease course is not necessarily prediction of treatment effect."

PRESERVE UNCERTAINTY

"The relationship to B17 response remains unestablished."

IDENTIFY THE NEXT EVIDENCE

"The next study must address the intended clinical relationship."

ASK ABOUT THE COHORT

"Which population and outcomes would establish the claimed relationship?"

CHECK THE CUTOFF

"What supports the proposed decision threshold?"

COMPARE SAMPLES CAREFULLY

"Performance in plasma does not automatically transfer to another sample matrix."

CLOSE THE SLIDE REVIEW

"Keep the measurement result and the proposed clinical use in separate statements."

Sensitive

Specify analytical detection or diagnostic classification.

Research units

Do not present an invented study scale as a clinical reference range.

Clinical utility

A relationship may be real without yet showing that using the test improves a decision.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which description matches the supplied evidence?

- A** The protein is qualified for every clinical use.
- B** A result above ten guarantees benefit.
- C** The test met analytical criteria from 2 to 20 research units; prediction of B17 response is unestablished.
- D** The test selects all B17 responders reliably.

2. Why is a predictive label premature?

- A** The analyte is measured in plasma.
- B** A measurement range was defined.
- C** The analytical criteria were predefined.
- D** No study has established differing B17 response associated with the marker.

3. A colleague chooses 11 as an eligibility cutoff because it is the midpoint of 2 to 20. What evidence is missing?

- A** Evidence that 11 lies inside the assessed measurement range
- B** Evidence that the midpoint of 2 and 20 equals 11
- C** Evidence that the analytical study used plasma samples
- D** Evidence supporting classification by that cutoff for the proposed B17 use

4. What would changing the sample from plasma require in the claim?

- A** Assuming identical performance because the analyte has the same name
- B** Calling the new matrix clinically validated without data
- C** Removing all information about sample conditions
- D** Appropriate evaluation of the changed matrix rather than automatic transfer of performance

Answers and explanations: [page 89](#).

Validated for which use?

E Conversations / turns 1-10

03

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

context of use / analyte / predictive biomarker / clinical validation / decision threshold / prognostic biomarker / analytical validation / matrix effect / clinical utility / Biomarker qualification

- 01 **Sara:** The slide says validated predictive biomarker and then proposes selecting participants by the protein value. Can you tell me which part of that statement our study establishes?
- 02 **Noah:** It establishes the assessed ¹ _____ scope: the test met predefined measurement criteria from two to twenty research units in the plasma conditions we evaluated.
- 03 **Sara:** That is useful, but it does not tell us whether a participant with a particular value is more likely to respond to B17 than another participant.
- 04 **Noah:** Correct. We have not established ² _____ for that proposed relationship. The measurement study did not test whether the marker identifies differences in response to this candidate.
- 05 **Sara:** We should state the planned decision explicitly, then. The proposed use is selecting participants for the B17 program, not simply reporting protein concentration in a research sample.
- 06 **Noah:** That is the ³ _____ that needs supporting evidence. A reliable measurement can be necessary without being sufficient for the clinical decision being proposed.
- 07 **Sara:** What should replace validated predictive biomarker in the current deck? I do not want to lose the analytical achievement while correcting the clinical overstatement.
- 08 **Noah:** Name the ⁴ _____, tested matrix, and range, then state that response prediction remains unestablished. We can report the achievement accurately without borrowing a broader label.
- 09 **Sara:** Why did the team choose ten units as the cutoff? Being inside the measurement range does not tell us who is likely to respond to B17.
- 10 **Noah:** A ⁵ _____ needs a justified relationship to the intended decision. The assay's ability to measure a value does not tell us what that value means for response.

The conversation, continued

E Conversations / turns 11-20

03

Continue with the word bank on page 28.

- 11 **Sara:** Suppose another study later associates the protein with faster disease progression. Would that automatically make it predictive for B17?
- 12 **Noah:** No. A ⁶ _____ concerns the likely clinical course. That association would not by itself establish that the marker identifies a differing effect of this intervention.
- 13 **Sara:** The distinction needs to appear in our terminology sheet. People use predict in everyday English for almost any association with a later event.
- 14 **Noah:** In this development claim, a ⁷ _____ relates to the effect of a particular intervention. We should define the intended scientific meaning rather than assume the everyday word settles it.
- 15 **Sara:** Another team wants to transfer the assay to a different sample type. They believe the protein name being unchanged means the measurement evidence transfers automatically.
- 16 **Noah:** We need to evaluate possible ⁸ _____ and the changed conditions. The same analyte in another sample environment is not automatically covered by the plasma performance study.
- 17 **Sara:** And an internal study meeting its criteria is not the same as receiving formal acceptance for every possible regulatory use. We should remove qualified from that sentence too.
- 18 **Noah:** Yes. ⁹ _____ has an applicable process and defined scope. We should not imply that such a determination exists when the evidence described here does not establish one.
- 19 **Sara:** I will remove validated predictive biomarker and the implied selection rule. Please give me wording that preserves the analytical result and names the clinical question still open.
- 20 **Noah:** Then later we can also assess ¹⁰ _____ for the relevant decision. For now, accurate measurement is a defined achievement, and the clinical application remains a separate evidence question.

ANSWERS / EXPLANATIONS: PAGE 90

1. analytical validation | 2. clinical validation | 3. context of use | 4. analyte | 5. decision threshold | 6. prognostic biomarker | 7. predictive biomarker | 8. matrix effect | 9. Biomarker qualification | 10. clinical utility

Measurement is not selection

F Say it

03

A test measures a protein adequately from 5 to 50 study units. No relationship to treatment response has been studied. A colleague proposes using 25 as an eligibility cutoff.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

selection / unstudied / threshold / range

Scientist: "Five to fifty is the assessed measurement ¹ _____."

Colleague: "Twenty-five is a proposed decision ² _____."

Scientist: "Its relationship to response is ³ _____."

Colleague: "The analytical result does not authorize participant ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 90

1. range | 2. threshold | 3. unstudied | 4. selection

Rehearse with a fixed script

1. Read the biomarker discussion. Keep the 2-to-20 range and tested plasma conditions with the analytical result.
2. Swap roles. Distinguish measuring the protein from predicting B17 response; do not create an eligibility rule.
3. Read the corrected threshold transfer. Separate the 5-to-50 measurement range from the proposed cutoff of 25.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A model effect is not proof of human benefit

04

A Read the situation

Summarize a nonclinical finding with its design and limitations, then identify missing information for the next review.

A fictional animal-model report compares 12 treated animals with 12 controls and reports a 20% lower mean model-endpoint value at one tested exposure. Groups were randomized; assessor blinding is not documented in the supplied summary. No uncertainty interval, exposure-response series, or human outcome data are provided. Scientist Amir and development lead Zoe review a slide calling the result proof of human benefit. They need a precise nonclinical summary and the missing study details, not an invented clinical conclusion.

PEOPLE / Amir: Scientist | Zoe: Development lead

REPORT WHAT THE STUDY CAN CARRY

Development discussions reward concise conclusions, but a short sentence should still name the model, comparison, and limits. Separate a positive result from a claim about people. Asking for missing design information strengthens the review without implying that the study was necessarily conducted badly.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the reported comparison?

- A** Identical results across every tested exposure
- B** A completed clinical trial in 24 patients
- C** A 20% lower mean model-endpoint value in 12 treated animals versus 12 controls at one exposure
- D** A 20% improvement in human survival

2. What is known about blinding?

- A** The assessors were definitely unblinded.
- B** Randomization automatically establishes blinding.
- C** It is not documented in the supplied summary.
- D** The assessors were definitely blinded.

3. Which conclusion is unsupported?

- A** The groups were randomized.
- B** One exposure was tested.
- C** Further study details are needed.
- D** Human benefit has been proved.

Answers and explanations: page 91.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

preclinical package

The body of nonclinical evidence supporting a development review.

IN USE assemble a preclinical package

model relevance

How an experimental system relates to the biological question of interest.

IN USE assess model relevance

sex as a biological variable

Consideration of sex-related differences in design, analysis, and reporting.

IN USE address sex as a biological variable

randomization

Allocation by a chance-based process to reduce systematic selection differences.

IN USE document randomization

blinding

Keeping relevant group or treatment information hidden from specified people.

IN USE verify assessor blinding

sample size

The number of experimental units in a defined analysis.

IN USE justify the sample size

experimental unit

The independently assigned unit underlying an experimental comparison.

IN USE identify the experimental unit

pharmacokinetics

The time course of a substance's absorption, distribution, metabolism, and elimination.

IN USE characterize pharmacokinetics

pharmacodynamics

The biological effects of an intervention and their relationship to exposure.

IN USE evaluate pharmacodynamics

exposure

The amount or concentration reaching a biological system over time.

IN USE characterize exposure

exposure-response relationship

The relationship between measured exposure and an observed effect.

IN USE examine the exposure-response relationship

bioavailability

Rate and extent of systemic availability of an active substance.

IN USE assess bioavailability

Precision check

A 20% lower model-endpoint mean is not automatically a 20% clinical benefit or a statistically established effect. State the comparison and missing uncertainty information. Randomization and blinding are distinct design features; one does not establish the other.

The vocabulary that does the work

B Find the words / 2 of 2

04

Read the definitions and say the common word combinations aloud.

biodistribution

The pattern of a substance's presence across tissues or body compartments.

IN USE characterize biodistribution

efficacy endpoint

A defined measurement used to assess the intended effect in a study.

IN USE specify the efficacy endpoint

tolerability

How adverse effects affect the ability to tolerate an intervention.

IN USE report tolerability findings

toxicology

Study of harmful effects of substances under specified conditions.

IN USE review toxicology data

NOAEL

No-observed-adverse-effect level under the conditions of a defined study.

IN USE interpret the NOAEL in context

safety margin

A comparison of relevant exposure or dose levels used in safety assessment.

IN USE justify the safety-margin comparison

good laboratory practice

A formal quality system for applicable nonclinical safety studies.

IN USE verify good laboratory practice status

dose-response relationship

The relationship between administered dose and a measured effect.

IN USE distinguish dose-response from exposure-response

translational uncertainty

Uncertainty in applying findings from one biological setting to another.

IN USE communicate translational uncertainty

study report

The documented methods, results, and interpretation of a study.

IN USE review the full study report

data provenance

The origin and processing history of data used in a result.

IN USE preserve data provenance

uncertainty interval

A stated range expressing uncertainty under a defined analytical method.

IN USE report the uncertainty interval

Precision check

Dose is what is administered; exposure concerns what reaches the biological system over time. PK means pharmacokinetics and PD means pharmacodynamics. A result at one exposure does not describe an exposure-response relationship or establish a human safety margin.

Phrases you can actually use

C Notice the language / 1 of 2

NAME THE MODEL RESULT

"The treated group had a 20% lower mean model-endpoint value than controls."

IDENTIFY THE TESTED CONDITION

"The result comes from one tested exposure."

SEPARATE DESIGN FEATURES

"Randomization is documented; assessor blinding is not stated in this summary."

ASK ABOUT UNCERTAINTY

"Which interval or analysis describes uncertainty around the reported difference?"

STATE THE GROUPS

"There were twelve treated animals and twelve controls."

KEEP THE SPECIES BOUNDARY

"Human benefit has not been established by these data."

REQUEST THE SOURCE

"Please provide the full report and the relevant analysis details."

AVOID A SIGNIFICANCE CLAIM

"The supplied percentage alone does not establish statistical significance."

Twenty-percent lower

Specify which endpoint, comparison, and mean; do not translate it into a patient claim.

Significant

Distinguish statistical evidence from an everyday judgment of importance.

Not documented

Does not automatically mean not performed.

Phrases you can actually use

C Notice the language / 2 of 2

NAME THE MISSING RELATIONSHIP

"We do not yet have an exposure-response series."

CLARIFY PK AND PD

"Distinguish the exposure measurements from the biological response."

PRESERVE UNIT DEFINITIONS

"Identify the experimental units behind the reported sample size."

AVOID AN INVENTED MARGIN

"We cannot calculate a human safety margin from these facts alone."

KEEP SAFETY SEPARATE

"An efficacy-endpoint result does not establish the complete safety profile."

CHECK STUDY STATUS

"Do not infer formal GLP compliance from the word preclinical."

LIMIT THE TRANSLATION

"This supports further investigation within a defined development question."

CLOSE THE PACKAGE REVIEW

"Summarize the finding, its limits, and the specific evidence still required."

PK/PD

Expand the terms for mixed audiences before using shorthand.

GLP

A defined quality framework, not a general compliment about laboratory work.

Safe

Specify the evidence and setting; one positive endpoint does not establish an unrestricted safety claim.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which sentence best represents the result?

- A** At one exposure, the treated animal group had a 20% lower mean model endpoint than controls; human benefit is unestablished.
- B** Humans improve by 20% with this intervention.
- C** The treatment is proven safe at every exposure.
- D** Blinded human efficacy was confirmed.

2. What does the missing blinding information mean?

- A** The review needs clarification from the study record.
- B** Randomization proves the assessors were blinded.
- C** The study definitely had no blinding.
- D** The endpoint can be described as independently replicated in humans.

3. Which addition would directly clarify uncertainty around the reported 20% model-endpoint difference?

- A** Confirmation that both groups contained twelve animals, without their results
- B** The reported randomization method alone, without variability or analysis
- C** The proposed human population, without additional model-study results
- D** The underlying group results and a justified uncertainty analysis

4. Which distinction is accurate?

- A** Administered dose and measured exposure are related but not identical concepts.
- B** One tested exposure establishes every dose-response relationship.
- C** An efficacy endpoint is the complete toxicology package.
- D** Formal GLP status follows automatically from using animals.

Answers and explanations: [page 91](#).

A model effect is not proof of human benefit

E Conversations / turns 1-10

04

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

translational uncertainty / Randomization / exposure-response relationship / study report / uncertainty interval / pharmacokinetics / efficacy endpoint / model relevance / safety margin / toxicology

- 01 **Zoe:** The development slide says this is proof of human benefit. The reported model effect is encouraging, but I need a sentence that will survive a detailed package review.
- 02 **Amir:** The ¹ _____ mean was twenty percent lower in twelve treated animals than in twelve controls. That is the comparison the study summary actually reports.
- 03 **Zoe:** We should say it was an animal model in the headline, not leave the audience to find that in a footnote after hearing a clinical claim.
- 04 **Amir:** Agreed. The ² _____ needs explanation, but relevance does not make the model identical to human disease. No human outcome data are included in this package.
- 05 **Zoe:** I see randomized allocation, but nothing about assessor blinding. Can you obtain that record before anyone adds blinded to the summary?
- 06 **Amir:** ³ _____ and assessor blinding are separate design features. The first is documented here; the second is not stated, so we need clarification rather than an assumption.
- 07 **Zoe:** I will request that detail from the study team. An omission in this summary does not prove the assessors knew the assignments, but it leaves the status unclear.
- 08 **Amir:** Ask for the relevant ⁴ _____ section and any supporting record. We should correct the summary from verified information, not fill the gap with the most favorable wording.
- 09 **Zoe:** The twenty-percent difference also appears without an interval. I cannot tell how much uncertainty surrounds the estimate from the headline alone.
- 10 **Amir:** We need the appropriate ⁵ _____ or other specified analysis and the underlying results. A percentage by itself does not establish statistical significance or the reliability of the estimate.

The conversation, continued

E Conversations / turns 11-20

04

Continue with the word bank on page 37.

- 11 **Zoe:** The team tested one exposure. That can establish a result under that condition, but it cannot show the whole pattern across lower and higher exposures.
- 12 **Amir:** Correct. We do not yet have an ⁶ _____. Keep that limitation beside the result so the development discussion does not assume a series that was never performed.
- 13 **Zoe:** We also need to separate an administered dose from what reaches the relevant biological system. Those terms are being used interchangeably in the notes.
- 14 **Amir:** The ⁷ _____ information addresses the disposition and exposure over time. The biological effect belongs to the pharmacodynamic question; neither can be replaced by a vague statement that the dose worked.
- 15 **Zoe:** What about the sentence saying the favorable endpoint establishes safety? I assume the full safety question requires different information and appropriate professional review.
- 16 **Amir:** Yes. Review the relevant ⁸ _____ and other safety evidence separately. We should not infer a complete safety profile from an endpoint designed to assess the intended effect.
- 17 **Zoe:** Nor should we invent a human safety margin from this one model result. We do not have the necessary comparison or its scientific justification.
- 18 **Amir:** Exactly. Any ⁹ _____ must state the compared quantities and supporting basis. A reassuring label cannot supply the missing exposure information or translation rationale.
- 19 **Zoe:** I will keep the result explicitly nonclinical and request the missing uncertainty and exposure information. Please also confirm the blinding status from the study records.
- 20 **Amir:** That communicates ¹⁰ _____ without discarding the finding. The result can justify further investigation while remaining distinct from proof that people will benefit.

ANSWERS / EXPLANATIONS: PAGE 92

1. efficacy endpoint | 2. model relevance | 3. Randomization | 4. study report | 5. uncertainty interval | 6. exposure-response relationship | 7. pharmacokinetics | 8. toxicology | 9. safety margin | 10. translational uncertainty

An omission is not a finding

F Say it

A study summary documents random allocation but does not describe assessor blinding. A colleague proposes writing either fully blinded or definitely unblinded without checking the report.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

record / verification / unstated / allocation

Scientist: "Random ¹ _____ is documented."

Reviewer: "Assessor blinding remains ² _____."

Scientist: "We need the relevant study ³ _____."

Reviewer: "Neither proposed label is supported without ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 92

1. allocation | 2. unstated | 3. record | 4. verification

Rehearse with a fixed script

1. Read the preclinical review. Preserve twelve treated animals, twelve controls, one exposure, and the 20% model-endpoint difference.
2. Swap roles. State blinding not documented rather than blinded or unblinded; request the relevant study record.
3. Read the corrected omission transfer. Keep random allocation distinct from unverified assessor blinding.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Two small runs do not prove the large process

A Read the situation

05

Distinguish titer, recovered quantity, product quality, and scale-up evidence in a process-transfer discussion.

Process scientist Hana and manufacturing lead Ben review two fictional 1-liter runs after a process transfer. Both achieved a harvest titer of 0.8 grams per liter. A 200-liter run has not been performed. The slide promises 160 grams of finished acceptable product by multiplying titer by volume. Downstream recovery and relevant quality results at the larger scale are unestablished. The team must retain the small-run result while correcting the forecast and defining the missing scale-up evidence.

PEOPLE / Hana: Process scientist | Ben: Manufacturing lead

A CALCULATION CAN BE CORRECT AND THE PROMISE WRONG

Operational audiences need quantities and delivery expectations, but an arithmetic projection must keep its assumptions visible. Clarify the process stage, recovery basis, and quality status. A useful estimate is conditional; it should not quietly become a commitment to release finished material.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What was demonstrated?

- | | |
|---|---|
| A A successful 200-liter run | B Complete quality comparability at all scales |
| C Harvest titer of 0.8 g/L in two 1-liter runs | D Release of 160 grams of finished product |

2. What does 0.8 g/L multiplied by 200 liters give under the stated assumption?

- | | |
|---|--|
| A A 160% recovery rate | B A validated clinical dose |
| C A hypothetical 160 grams at harvest before downstream losses | D A guaranteed 160 grams of released finished product |

3. Which evidence remains missing?

- | | |
|--|--|
| A Whether two small runs occurred | B The small-run titer value |
| C The proposed large-run volume | D Larger-scale performance, downstream recovery, and relevant quality results |

Answers and explanations: page 93.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

CMC

Chemistry, manufacturing, and controls: the product and process quality package.

IN USE review the CMC package

technology transfer

Movement of process knowledge and capability between teams or sites.

IN USE execute technology transfer

upstream processing

The production stages that generate the biological material.

IN USE characterize upstream processing

downstream processing

Stages that recover and purify material after production.

IN USE evaluate downstream processing

titer

Product concentration at a specified process stage.

IN USE report harvest titer

yield

Output relative to a stated input or theoretical basis, with context required.

IN USE define the yield basis

recovery

The proportion of material retained through a specified process step.

IN USE calculate downstream recovery

purity

The proportion or profile of intended material relative to impurities.

IN USE assess product purity

potency

A measure of biological activity relevant to the product's function.

IN USE evaluate product potency

aggregation

Association of molecules into larger assemblies.

IN USE monitor product aggregation

critical quality attribute

A property needing control within appropriate limits to ensure desired quality.

IN USE identify critical quality attributes

critical process parameter

A process variable whose variability can affect a critical quality attribute.

IN USE assess critical process parameters

Precision check

At the same assumed titer, 0.8 grams per liter times 200 liters gives 160 grams at harvest. It does not establish downstream recovery, final purity, potency, or release. Titer is a concentration; always define the basis when using yield.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

process parameter

A measurable or controlled operating variable in a process.

IN USE document process parameters

process characterization

Study of how process conditions affect performance and quality.

IN USE plan process characterization

scale-up

Increasing the size or production capacity of a process.

IN USE evaluate scale-up risks

scale-down model

A smaller model intended to represent relevant larger-process behavior.

IN USE qualify a scale-down model

mass transfer

Movement of a substance between regions or phases of a system.

IN USE assess mass-transfer behavior

mixing

Distribution of material within a process vessel or system.

IN USE characterize mixing

hold time

The time material remains at a specified stage or condition.

IN USE justify hold times

bioburden

The level of viable microorganisms associated with a material or process.

IN USE monitor bioburden

comparability

Assessment that a change does not adversely affect relevant product quality, safety, or efficacy.

IN USE establish a comparability strategy

change control

Formal review and management of a proposed change.

IN USE follow change control

process validation

Documented evidence that a process can consistently deliver the intended quality.

IN USE plan process validation

batch record

The record of materials, steps, observations, and events for a batch.

IN USE review the batch record

Precision check

CMC means chemistry, manufacturing, and controls; CQA means critical quality attribute; CPP means critical process parameter. Matching one output does not establish complete product comparability. Assess relevant quality properties and process changes through the applicable development and quality framework.

Phrases you can actually use

C Notice the language / 1 of 2

STATE THE OBSERVED SCALE

"Both results came from one-liter runs."

SHOW THE ARITHMETIC

"At the same titer, two hundred liters would contain one hundred sixty grams at harvest."

SEPARATE RECOVERY

"Downstream losses are not included in that calculation."

CLARIFY THE ACRONYM

"CMC covers chemistry, manufacturing, and controls."

NAME THE STAGE

"The measured value is harvest titer, not released final quantity."

EXPOSE THE ASSUMPTION

"The larger process has not yet demonstrated that titer."

KEEP QUALITY VISIBLE

"Quantity alone does not establish purity or potency."

ASK ABOUT ATTRIBUTES

"Which quality attributes need comparison after this change?"

Grams per liter

Concentration, not total quantity.

Harvest versus finished

Material at harvest has not necessarily passed through purification and release.

Recovery

Specify the process stage and input basis.

Phrases you can actually use

C Notice the language / 2 of 2

AVOID PREMATURE VALIDATION

"Two small runs do not establish the complete larger-scale process."

IDENTIFY THE EVIDENCE

"We need relevant larger-scale performance and quality data."

DEFINE THE MODEL

"Which aspects of the large process does the small model represent?"

SEPARATE RELEASE

"A projected quantity is not a release decision."

CHECK PROCESS BEHAVIOR

"Mixing and mass transfer may differ with scale."

RESPECT CHANGE REVIEW

"Route the process change through the appropriate quality review."

QUALIFY THE FORECAST

"Present this as a conditional harvest estimate."

CLOSE THE TRANSFER

"Document the observed performance, open risks, and next agreed evaluation."

Same titer

Does not establish the same impurity profile or biological activity.

Scale-up

A larger vessel does not guarantee proportionally identical performance.

Validated

State the process, scope, and supporting evidence rather than use it as a general success label.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which forecast is appropriately qualified?

- A** The larger process is validated without being run.
- B** If 0.8 g/L holds at 200 liters, harvest quantity would be 160 grams before recovery and quality evaluation.
- C** Two small runs guarantee 160 grams of released product.
- D** Titer alone proves complete comparability.

2. If a separate hypothetical process retains 70% of 160 grams, how much remains after that step?

- A** 112 grams
- B** 230 grams
- C** 70 grams
- D** 160 grams

3. Which item is a quality question rather than simply a quantity calculation?

- A** Whether two one-liter runs were reported
- B** Whether relevant potency and impurity attributes remain appropriate
- C** Whether 0.8 times 200 equals 160
- D** Whether the proposed volume is 200 liters

4. Both one-liter runs have the same titer. Which proposed summary overstates the demonstrated scope?

- A** Larger-scale recovery and quality evidence remain outstanding.
- B** The 160-gram figure assumes the titer holds at 200 liters.
- C** Matching titer establishes larger-scale quality comparability.
- D** Matching titer is one result from the small-run process.

Answers and explanations: [page 93](#).

Two small runs do not prove the large process

E Conversations / turns 1-10

05

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

scale-down model / titer / Potency / scale-up / mass transfer / critical quality attribute / change control / CMC / comparability / recovery

- 01 **Ben:** I have a production slide promising 160 grams of acceptable finished product at 200 liters. That comes from the two one-liter runs. Can we support it?
- 02 **Hana:** The measured ¹ _____ was 0.8 grams per liter at harvest. We have not run 200 liters, and finished material is a different stage.
- 03 **Ben:** Then 160 is a correct multiplication with an unproven scale assumption. I also mislabeled where in the process that quantity would exist.
- 04 **Hana:** Yes. The ² _____ still needs evidence. Write conditional harvest estimate, not finished-product commitment, so the receiving team can see both limits.
- 05 **Ben:** What would change the amount after harvest? The table currently carries all 160 grams straight through purification without showing any loss.
- 06 **Hana:** It omits downstream ³ _____. We need the retained fraction for the relevant stages, not an implicit assumption that all harvested material reaches the end.
- 07 **Ben:** If a later run did match the concentration, could I at least use quality confirmed? The slide needs a short status for the review.
- 08 **Hana:** Matching concentration does not verify every ⁴ _____. A short status still has to preserve the relevant purity, activity, and other quality questions.
- 09 **Ben:** So an equal number of grams does not necessarily mean an equal biological effect. That distinction needs to survive when this becomes a commercial forecast.
- 10 **Hana:** Correct. ⁵ _____ addresses relevant biological activity. The concentration alone does not establish it, just as it does not describe the full impurity profile.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 46.

- 11 **Ben:** We are also changing vessel size. I do not want the scale discussion reduced to multiplying every small-run setting by two hundred.
- 12 **Hana:** The review needs relevant ⁶ _____ and mixing behavior. Larger volume can change process conditions; proportional arithmetic does not demonstrate equivalent performance.
- 13 **Ben:** Can we still use the small system to investigate risks? I do not want to discard two useful runs because they answer a narrower question.
- 14 **Hana:** We can establish what a ⁷ _____ represents. Its value depends on the relevant behavior demonstrated, not merely on using a smaller vessel.
- 15 **Ben:** I will attach the process versions and batch records to the transfer pack. Which route keeps the proposed changes and their review traceable?
- 16 **Hana:** Use ⁸ _____ and the applicable transfer process. The receiving team needs the actual settings, changes, and supporting records, not just the headline.
- 17 **Ben:** One last wording point: two matching titers do not support saying comparability demonstrated. We still lack the relevant larger-scale quality results.
- 18 **Hana:** Exactly. Establishing ⁹ _____ requires examining the consequences for the product. A single matching output is not the whole assessment.
- 19 **Ben:** The revised slide will say two one-liter runs at 0.8 grams per liter; 160 grams is hypothetical harvest output before recovery and quality evaluation.
- 20 **Hana:** That is a usable ¹⁰ _____ update. Keep finished acceptable output unconfirmed and name the remaining evidence needed before making that commitment.

ANSWERS / EXPLANATIONS: PAGE 94

1. titer | 2. scale-up | 3. recovery | 4. critical quality attribute | 5. Potency | 6. mass transfer | 7. scale-down model | 8. change control | 9. comparability | 10. CMC

Concentration is not the final quantity

F Say it

A hypothetical harvest contains 0.5 grams per liter in 100 liters. A stated downstream step retains 80% of the harvest material. No release decision is supplied.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

separate / fifty / forty / release

Scientist: "The harvest contains ¹ _____ grams."

Reviewer: "At eighty-percent recovery, ² _____ grams remain."

Scientist: "That is a quantity calculation, not a ³ _____ decision."

Reviewer: "Keep the relevant product-quality evidence ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 94

1. fifty | 2. forty | 3. release | 4. separate

Rehearse with a fixed script

1. Read the scale-up exchange. Distinguish observed one-liter titer from assumed 200-liter performance and harvest from finished output.

2. Swap roles. State 0.8 times 200 equals 160 grams before downstream recovery; do not claim release.

3. Read the corrected transfer. Say fifty grams at harvest and forty after the stated step, with quality assessment separate.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A confidentiality agreement is not a material license

A Read the situation

Identify unresolved material-use and publication terms while keeping ownership, permission, and freedom to operate distinct.

06

Two fictional research partners plan to exchange a nonclinical reference protein for three named bench assessments. A confidentiality agreement is signed, but the material-transfer terms remain a draft. Permitted use, onward transfer, publication review, and rights in new findings are unresolved. Scientist Julia asks counsel Ravi whether shipment can proceed and whether owning a related patent settles third-party rights. Their discussion identifies questions for the authorized review; it does not interpret a real agreement or approve a transfer.

PEOPLE / Julia: Scientist | Ravi: Counsel

A SHARED SCIENTIFIC AIM DOES NOT SETTLE THE AGREEMENT

Researchers may use collaborate to mean anything from a limited evaluation to joint development. Define the intended work and the relevant rights before relying on that word. Asking about publication or ownership early can protect the relationship by exposing different expectations before materials move.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which agreement is signed?

- | | |
|---|--|
| A The final material-transfer agreement | B A worldwide exclusive development license |
| C A completed assignment of every future invention | D The confidentiality agreement |

2. Which issue remains unresolved?

- | | |
|--|--|
| A Whether counsel has been asked for review | B Permitted use and publication review, among other material-transfer terms |
| C Whether a reference protein is proposed | D Whether three bench assessments are contemplated |

3. What does ownership of a related patent establish about third-party rights here?

- | | |
|--|--|
| A It replaces the material-transfer review. | B It does not by itself establish freedom to operate. |
| C It automatically removes every third-party right. | D It authorizes every recipient use worldwide. |

Answers and explanations: page 95.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

intellectual property

Legally recognized rights in inventions, works, marks, or other protected subject matter.

IN USE identify relevant intellectual property

background IP

Intellectual property brought into a collaboration, as defined by the agreement.

IN USE define background IP

foreground IP

New intellectual property arising from the collaboration under its agreed definitions.

IN USE address foreground IP

inventorship

Legal identification of the people who contributed to a claimed invention.

IN USE assess inventorship

ownership

The legal holding of a right or asset.

IN USE clarify ownership

assignment

Transfer of ownership of specified rights.

IN USE distinguish an assignment from a license

license

Permission to exercise specified rights under stated terms.

IN USE negotiate a license

field of use

The defined application area within which permission applies.

IN USE limit the field of use

territory

The geographic scope of a right or permission.

IN USE specify the territory

exclusive license

Permission with defined exclusivity within its agreed scope.

IN USE define exclusive-license scope

nonexclusive license

Permission that does not generally prevent granting similar rights to others.

IN USE discuss a nonexclusive license

confidentiality obligation

A duty concerning protection and permitted disclosure of information.

IN USE comply with confidentiality obligations

Precision check

NDA means nondisclosure agreement; MTA means material transfer agreement; FTO means freedom to operate. Confidentiality, material permission, ownership, and third-party rights are different questions. The effect of any document depends on its actual language, scope, law, and circumstances.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

nondisclosure agreement

An agreement governing specified confidential information.

IN USE review a nondisclosure agreement

material transfer agreement

An agreement governing the transfer and use of tangible research material.

IN USE finalize a material transfer agreement

permitted use

The activities authorized for the material or rights under the agreement.

IN USE define permitted use

onward transfer

Passing material or rights to another recipient.

IN USE address onward transfer

publication review

An agreed process for reviewing proposed public disclosure.

IN USE specify publication review

publication embargo

A stated restriction or delay on publication under applicable terms.

IN USE define a publication embargo

patent claim

The legal wording defining the scope of protection sought or granted.

IN USE review relevant patent claims

patentability

Whether an invention meets the applicable criteria for patent protection.

IN USE assess patentability

freedom to operate

Whether proposed acts can proceed without infringing relevant third-party rights.

IN USE obtain a scoped freedom-to-operate review

know-how

Practical technical information or expertise, sometimes protected as confidential information.

IN USE identify transferred know-how

third-party rights

Rights held by someone outside the parties to the proposed arrangement.

IN USE assess third-party rights

governing agreement

The contract whose applicable terms control the relevant relationship or activity.

IN USE identify the governing agreement

Precision check

A patent generally provides a right to exclude specified acts by others, not an automatic right to practice every related activity. Patentability and freedom to operate therefore ask different questions. Obtain appropriately scoped professional review rather than infer clearance from ownership.

Phrases you can actually use

C Notice the language / 1 of 2

IDENTIFY THE SIGNED DOCUMENT

"The confidentiality agreement is signed; the material-transfer terms are not final."

SEPARATE PERMISSIONS

"Permission to receive information does not necessarily authorize use of the material."

CLARIFY NEW FINDINGS

"How will rights in results and new inventions be addressed?"

PROTECT PUBLICATION

"What review process and time limits apply before public disclosure?"

DEFINE THE ACTIVITY

"The proposed use is three named nonclinical bench assessments."

CHECK ONWARD TRANSFER

"May the recipient send the material to another laboratory?"

SEPARATE INVENTORSHIP

"Inventorship and ownership should not be treated as interchangeable labels."

AVOID AN UNLIMITED DELAY

"Publication review needs defined terms, not an unstated permanent veto."

06

Own versus use

Holding one right and having permission for a proposed activity are different questions.

NDA versus MTA

The labels indicate different purposes; read the actual terms rather than assume either is comprehensive.

Research only

Define the permitted research and any restrictions instead of relying on a broad phrase.

Phrases you can actually use

C Notice the language / 2 of 2

NAME EXISTING RIGHTS

"Identify the background IP each party brings."

ASK ABOUT THIRD PARTIES

"Which relevant rights are held outside these two parties?"

AVOID A PATENT SHORTCUT

"Owning our patent does not automatically establish freedom to operate."

RECORD THE OPEN TERMS

"Keep permitted use and publication questions visible in the draft."

DEFINE THE NEW WORK

"Distinguish collaboration results from pre-existing know-how."

SCOPE FTO

"The review needs the proposed acts, territories, and relevant time."

RESERVE AUTHORIZATION

"Shipment needs the appropriate completed review and authorization."

CLOSE ACCURATELY

"Scientific interest is aligned; the transfer terms remain unresolved."

06

Publication review

Specify purpose, timing, and consequences through the actual agreement.

Inventor versus owner

The legal concepts differ and may involve different people or entities.

Cleared

State the reviewed activity and scope; do not present a limited review as universal permission.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which response best fits the shipment question?

- A** Shipment proves the terms were accepted in advance.
- B** The MTA terms remain unresolved, so seek the required review and authorization before treating shipment as approved.
- C** The signed NDA automatically approves every material use.
- D** A common scientific goal replaces contract review.

2. Which review question concerns permitted use rather than a different transfer term?

- A** How long may the provider review a proposed manuscript?
- B** Who owns an invention arising from the collaboration?
- C** Which named bench assessments may the recipient perform with the protein?
- D** May the recipient send the protein to a third laboratory?

3. Why does a related owned patent not settle FTO?

- A** Patent ownership always means the patent is invalid.
- B** Every research material is automatically unprotected.
- C** An MTA can never contain use terms.
- D** Other parties may hold relevant rights affecting the proposed acts.

4. Which statement preserves the publication issue?

- A** A planned manuscript proves ownership of all inventions.
- B** The publication-review process and timing remain to be agreed.
- C** Every result may be published immediately because an NDA exists.
- D** Publication is forbidden forever regardless of the draft.

Answers and explanations: [page 95](#).

A confidentiality agreement is not a material license

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

06

WORD + PHRASE BANK

permitted use / Inventorship / material transfer agreement / background IP / governing agreement / nondisclosure agreement / publication review / third-party rights / freedom to operate / know-how

- 01 **Julia:** The partners agree on the science, and the confidentiality agreement is signed. Can I arrange shipment of the reference protein for the three bench assessments now?
- 02 **Ravi:** The ¹ _____ is still a draft. We need the relevant terms reviewed and authorized before treating the transfer as approved.
- 03 **Julia:** I had assumed the signed confidentiality document covered the exchange. It lets us discuss the work, but that may be different from permission to use the material.
- 04 **Ravi:** Exactly. A ² _____ does not automatically answer every material-use question. We must examine the actual documents rather than assume the title grants whatever the project needs.
- 05 **Julia:** The intended work is limited to three named assessments. The recipient also mentioned asking a separate laboratory to help, which is not addressed in the draft.
- 06 **Ravi:** Define the ³ _____ precisely and address any onward transfer. The scientific plan should match the scope being reviewed, including which organizations would actually handle the material.
- 07 **Julia:** We also need to clarify what happens to discoveries made during the evaluation. Each team is bringing existing methods and information to the work.
- 08 **Ravi:** Identify the ⁴ _____ separately from new results. The definitions should let both parties distinguish their existing rights from what the collaboration may generate.
- 09 **Julia:** If a new invention results, naming the people who contributed is not necessarily the same as deciding which organization will own the resulting rights.
- 10 **Ravi:** Correct. ⁵ _____ and ownership are different legal questions. We should not resolve either casually in an email saying that every new idea belongs to the team.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 55.

06

- 11 **Julia:** The recipient wants to publish. Can the draft specify how review would work, including timing, without implying either immediate unrestricted publication or an unlimited veto?
- 12 **Ravi:** Agree a defined ⁶ _____ process with the relevant purpose, timing, and consequences. Do not assume immediate unrestricted publication or an unlimited veto when neither term has been settled.
- 13 **Julia:** They also asked whether access to the material means they can use our practical methods in a later unrelated project. That sounds broader than this evaluation.
- 14 **Ravi:** Clarify any ⁷ _____ being supplied and the authorized uses. Physical possession of a reference protein does not automatically settle rights in every method discussed alongside it.
- 15 **Julia:** Our team owns a related patent. Some colleagues think that means we can tell the partner there are no intellectual-property obstacles to any proposed use.
- 16 **Ravi:** Ownership does not itself establish ⁸ _____. Other rights may matter, and the review needs the actual activity, territory, and relevant circumstances before anyone makes a clearance statement.
- 17 **Julia:** Then our own patentability work and a third-party-rights review should not be presented as the same assessment. They answer different questions for the collaboration.
- 18 **Ravi:** Yes. Examine relevant ⁹ _____ through appropriately scoped professional review. A right we hold is not a universal permission to practice every related technology or distribute every associated material.
- 19 **Julia:** I will hold shipment and send the exact assessment list and laboratory names for review. The teams agree on the science; the transfer terms still need resolution.
- 20 **Ravi:** That gives us a clear ¹⁰ _____ to finalize through the proper process. Once the terms and authorization are established, the team can act within that defined scope rather than rely on assumptions.

ANSWERS / EXPLANATIONS: PAGE 96

1. material transfer agreement | 2. nondisclosure agreement | 3. permitted use | 4. background IP | 5. Inventorship | 6. publication review | 7. know-how | 8. freedom to operate | 9. third-party rights | 10. governing agreement

Access to a file is not every right

F Say it

A partner may receive a confidential method summary for evaluation. No permission for commercial use or onward disclosure has been agreed in the supplied facts.

Complete the four gaps. Use each bank entry once.

06

WORD + PHRASE BANK

permitted / evaluation / terms / unresolved

Scientist: "The stated purpose is ¹ _____."

Partner: "Commercial use is not established as ² _____."

Scientist: "Onward disclosure also remains ³ _____."

Partner: "We need the actual governing ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 96

1. evaluation | 2. permitted | 3. unresolved | 4. terms

Rehearse with a fixed script

1. Read the transfer review. Distinguish the signed confidentiality agreement from the unfinished material-transfer terms.
2. Swap roles. Separate permitted use, onward transfer, publication review, and ownership; retain the unapproved shipment status.
3. Read the corrected file-access transfer. Keep evaluation permission distinct from commercial use and onward disclosure.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Six experiments, one unfinished milestone

A Read the situation

Connect completed work to a scientific funding criterion and explain a conditional cash-runway calculation.

Program lead Daria and finance lead Miles prepare a fictional board update. Four binding experiments and two functional runs are complete. The agreed next funding criterion requires three consistent independent functional runs; the third is pending. Unrestricted cash is \$12 million and the base-case net cash burn is \$1.2 million per month. A proposed \$0.3 million follow-up budget is not yet approved or included in that base case. The board needs milestone status and a bounded decision request, not a count presented as proof.

PEOPLE / Daria: Program lead | Miles: Finance lead

A BOARD UPDATE SHOULD CONNECT EVIDENCE AND CHOICE

Senior audiences often ask whether a program is de-risked. Identify which uncertainty has been reduced and which criterion remains open. A clear request names the proposed work and resources without presenting a favorable experimental outcome or future funding as guaranteed.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Why is the funding criterion not yet met?

A Only two of the required three independent functional runs are complete.

B No experiments of any kind have been completed.

C Four binding experiments automatically count as functional runs.

D Cash balance alone determines the scientific result.

2. What is the simple base-case runway at unchanged net burn?

A One hundred months

B Ten months

C Twelve months

D One month

3. How should the \$0.3 million proposal be described?

A A guaranteed investment return

B Proof that the functional criterion is met

C Additional proposed funding, not yet approved or included in the base case

D Cash already spent and reflected in the base case

Answers and explanations: page 97.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

milestone criterion

The evidence or condition required to reach a defined project milestone.

IN USE specify the milestone criterion

readout

The reporting of results from a study or experiment.

IN USE schedule a data readout

evidence package

The organized results and supporting records used for a decision.

IN USE assemble the evidence package

go/no-go decision

A defined decision to continue, stop, or redirect a program.

IN USE support a go/no-go decision

scientific risk

Uncertainty about whether the underlying biological or technical premise will hold.

IN USE reduce scientific risk

execution risk

Uncertainty about delivering the planned work as intended.

IN USE manage execution risk

critical path

The sequence of dependent tasks determining the earliest completion point.

IN USE identify the critical path

dependency

An input or task that another activity relies on.

IN USE make dependencies explicit

board ask

The specific decision or support requested from the board.

IN USE state the board ask

resource allocation

Assignment of money, people, or capacity to defined work.

IN USE justify resource allocation

unrestricted cash

Cash available for use without the specified restrictions of restricted funds.

IN USE report unrestricted cash

net cash burn

Net cash outflow over a stated period.

IN USE calculate monthly net cash burn

Precision check

Six completed experiments do not satisfy a criterion requiring three consistent independent functional runs when only two are complete. The category and quality of evidence matter. A milestone is not met merely because the activity count looks substantial.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

gross cash burn

Total cash outflow before offsetting inflows over a stated period.

IN USE distinguish gross cash burn

cash runway

The estimated period before available cash is exhausted under stated assumptions.

IN USE estimate cash runway

run rate

A current rate projected over another period under stated assumptions.

IN USE qualify the run-rate estimate

base case

The central planning scenario used for comparison.

IN USE update the base case

downside case

A planning scenario with less favorable specified assumptions.

IN USE test the downside case

budget variance

The difference between planned and actual spending on the same basis.

IN USE explain a budget variance

contingent funding

Funding dependent on specified future conditions.

IN USE identify contingent funding

financing round

A defined transaction or process to raise capital.

IN USE prepare for a financing round

dilution

Reduction in an existing owner's proportional interest following additional equity issuance.

IN USE explain potential dilution

inflection point

A development event expected to materially change a program's position or perceived value.

IN USE define the proposed inflection point

milestone payment

A payment triggered by a specified achievement under an agreement.

IN USE verify milestone-payment conditions

decision record

A record of the decision, basis, conditions, and responsible parties.

IN USE maintain a decision record

Precision check

The simple runway is \$12 million divided by \$1.2 million per month: ten months at unchanged net burn. This is a conditional illustration, not a forecast guarantee. New spending, timing, inflows, or restrictions require an updated cash model.

Phrases you can actually use

C Notice the language / 1 of 2

LEAD WITH THE DECISION

"We are asking for a bounded follow-up budget, not declaring the milestone complete."

NAME THE CRITERION

"The gate requires three consistent independent functional runs."

SEPARATE EVIDENCE TYPES

"Binding data support the program but do not replace the functional criterion."

CONNECT THE SPEND

"The proposed budget supports the next defined evidence question."

REPORT THE WORK

"Four binding experiments and two functional runs are complete."

STATE THE GAP

"The third functional run is pending."

AVOID TOTAL DE-RISKING

"One uncertainty may be reduced while other risks remain."

STATE APPROVAL STATUS

"The additional \$0.3 million has not yet been approved."

07

Completed

Name the work completed rather than imply every milestone is complete.

De-risked

Specify the risk reduced; do not imply that uncertainty disappeared.

Runway

State cash, burn basis, and assumptions.

Phrases you can actually use

C Notice the language / 2 of 2

SHOW THE RUNWAY BASIS

"At unchanged net burn, twelve divided by one point two gives ten months."

SEPARATE CASH AND SCIENCE

"Available cash does not establish a successful scientific result."

USE A DOWNSIDE CASE

"Show what changes if the result or timing is less favorable."

RECORD THE CONDITIONS

"Capture the board's actual decision and any conditions."

KEEP THE MODEL CURRENT

"The new proposal is outside the current base case."

NAME THE DEPENDENCY

"The funding decision depends on the evidence criterion, not only the calendar."

AVOID A FUNDING PROMISE

"A planned financing round is not cash already received."

CLOSE WITH THE NEXT READOUT

"Return with the third-run result and the updated cash implications."

Funded

Distinguish a proposal, approval, contractual commitment, and cash received.

Milestone

Use the agreed criterion, not a convenient substitute created for the update.

Base case

An unapproved new cost is not automatically already included.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which opening is most decision-ready?

- A** Two functional runs are complete; the third required by the funding criterion is pending, and we request \$0.3 million for the defined follow-up.
- B** Six experiments prove the milestone is met.
- C** The program is completely de-risked because the slide is finished.
- D** The cash balance guarantees a positive third result.

2. What makes the ten-month runway conditional?

- A** It proves future funding will arrive.
- B** It includes every unapproved future cost automatically.
- C** It depends only on the number of experiments completed.
- D** It assumes unchanged net burn and the stated available cash basis.

3. The board packet counts six completed experiments against the three-run gate. What is the reporting error?

- A** It treats the two functional runs as pending even though they are complete.
- B** It excludes a third completed functional run that is already in the record.
- C** It combines different evidence categories although the criterion specifies functional runs.
- D** It understates the completed activity because six is fewer than three.

4. What should happen if the new budget is approved?

- A** Keep calling it excluded from every forecast permanently.
- B** Treat approval as proof of scientific success.
- C** Add the budget amount to cash as though it were revenue.
- D** Update the cash model for the actual spending assumptions and record the decision.

Answers and explanations: [page 97](#).

Six experiments, one unfinished milestone

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

dependency / base case / decision record / downside case / cash runway / evidence package / board ask / scientific risk / contingent funding / milestone criterion

07

- 01 **Daria:** The update opens with six experiments completed. That is accurate as an activity count, but I am concerned it makes the next funding milestone sound finished.
- 02 **Miles:** State the ¹ _____ before the count. The agreed requirement is three consistent independent functional runs, and we currently have two with the third still pending.
- 03 **Daria:** Four of the six completed experiments are binding work. They support the program, but they do not replace the particular functional evidence required at this gate.
- 04 **Miles:** Exactly. The ² _____ should separate those categories. The board needs to see which result addresses which criterion, not infer progress from a single large total.
- 05 **Daria:** We also need to revise the phrase fully de-risked. The completed work may reduce some uncertainty, but the pending functional result could still affect the direction.
- 06 **Miles:** Name the remaining ³ _____. Do not turn progress into a claim that every uncertainty has disappeared, particularly when the agreed evidence condition remains open.
- 07 **Daria:** My request is the additional three-hundred-thousand-dollar budget for the defined follow-up. I am not asking for an unlimited commitment to every later development stage.
- 08 **Miles:** Make that the ⁴ _____. State the work it supports and its approval status. At present the amount is proposed, not approved or already incorporated into the base case.
- 09 **Daria:** The finance panel shows twelve million dollars of unrestricted cash and net burn of one point two million a month. That gives ten months under unchanged assumptions.
- 10 **Miles:** Yes, as a simple ⁵ _____ illustration. Put the assumptions next to the number so it is not heard as a guarantee that every future spending plan is already covered.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 64.

- 11 **Daria:** Show the proposed three hundred thousand separately. What happens to the forecast if the board approves it, given that the base case excludes it?
- 12 **Miles:** Update the ⁶ _____ if the board approves new spending, using the actual timing assumptions. Approval changes the plan; it does not make the original calculation timeless.
- 13 **Daria:** There is also a proposed financing round later. We should keep that distinct from cash already available rather than use it to make the runway look longer today.
- 14 **Miles:** Correct. Treat any ⁷ _____ according to its actual status. A plan to raise money is not a receipt, and an unmet scientific condition should remain visible.
- 15 **Daria:** For the schedule, the third functional run is the key unresolved item. The board should understand how its outcome connects to the next program decision.
- 16 **Miles:** Identify that ⁸ _____ explicitly. The calendar alone does not satisfy an evidence gate, and a completed run still needs its result evaluated against the criterion.
- 17 **Daria:** We can also show what happens if the result does not support continuation or if the timing moves. That makes the spending choice more concrete.
- 18 **Miles:** Use a stated ⁹ _____ without presenting it as a prediction. The purpose is to show which assumptions change the resource choice and when the board would revisit it.
- 19 **Daria:** I will put the third functional run and the budget request on the decision slide. Please add when the board will see the result and updated cash position.
- 20 **Miles:** Capture the actual approval or deferral in the ¹⁰ _____. Keep funding authorization, experiment completion, and a successful scientific result distinct, even on the same slide.

07

ANSWERS / EXPLANATIONS: PAGE 98

1. milestone criterion | 2. evidence package | 3. scientific risk | 4. board ask | 5. cash runway | 6. base case | 7. contingent funding | 8. dependency | 9. downside case | 10. decision record

New spending changes the assumptions

F Say it

A fictional program has \$9 million of available cash and a steady net burn of \$1.5 million per month. A new project cost is proposed but excluded from that base case.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

six / model / excluded / burn

Finance lead: "The simple base-case runway is ¹ _____ months."

Program lead: "That assumes an unchanged net ² _____."

Finance lead: "The new project cost is currently ³ _____."

Program lead: "Approval would require an updated cash ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 98

1. six | 2. burn | 3. excluded | 4. model

Rehearse with a fixed script

1. Read the board update. Report four binding experiments and two functional runs, with the third functional run pending.
2. Swap roles. State ten months at the base-case burn and the unapproved \$0.3 million outside that model.
3. Read the corrected cash transfer. Use nine divided by one point five: six months before any new project cost.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Define the collaboration before granting exclusivity

A Read the situation

Clarify the scope and conditions of a proposed partnership without confusing a discussion, an option, and a granted license.

Biotech partnering lead Nina meets prospective partner Tomas. The proposed first stage is a six-month nonclinical evaluation of two fictional targets, T2 and T5. Tomas requests worldwide exclusive rights across all platform applications before the collaboration scope is defined. No license, option, budget, or governance terms are approved. Nina is authorized to discuss a narrower proposal, not grant rights. They must identify the scope, evidence, decision process, and unresolved commercial terms for a reviewed term sheet.

PEOPLE / Nina: Biotech partnering lead | Tomas: Prospective partner

08

EXPLORE INTERESTS BEFORE ACCEPTING THE BROAD WORDING

A request for exclusivity may reflect a concern about investing resources while a competitor receives the same opportunity. Acknowledge that concern, then clarify the protected activity and duration. A narrower proposal can keep negotiation moving without implying that rights have already been granted.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the proposed first-stage work?

- A An approved clinical trial across all diseases
- C A six-month nonclinical evaluation of T2 and T5

- B Permanent assignment of the whole platform
- D Worldwide commercial development of every application

2. What authority does Nina have?

- A Discuss a narrower proposal, not grant rights
- C Approve all governance and budget terms

- B Execute every exclusive license immediately
- D Assign unrelated third-party patents

3. What remains unapproved?

- A The identity of the two proposed targets
- C License, option, budget, and governance terms

- B The proposed six-month evaluation duration
- D The fact that Tomas requested exclusivity

Answers and explanations: page 99.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

partnering thesis

The argument for why a collaboration could create value for both parties.

IN USE articulate the partnering thesis

strategic fit

Alignment between the collaboration and the parties' capabilities or goals.

IN USE assess strategic fit

evaluation license

Permission limited to a defined assessment purpose under agreed terms.

IN USE scope an evaluation license

option agreement

An agreement giving a defined opportunity to obtain specified rights on stated conditions.

IN USE negotiate an option agreement

option exercise

The act of using an option according to its agreed conditions.

IN USE define option-exercise requirements

license grant

The clause or act conveying specified permission under a license.

IN USE delimit the license grant

exclusivity scope

The defined activities, rights, territory, and period covered by exclusivity.

IN USE narrow the exclusivity scope

field restriction

A limitation to specified applications or uses.

IN USE define a field restriction

territorial restriction

A geographic limit on the relevant rights or activities.

IN USE clarify territorial restrictions

reserved rights

Rights explicitly retained rather than granted to the other party.

IN USE identify reserved rights

sublicensing

Granting another party rights under authority provided by a license.

IN USE address sublicensing permission

diligence obligation

An agreed requirement to make specified efforts or progress.

IN USE negotiate diligence obligations

Precision check

A proposal to discuss exclusivity does not grant it. A term sheet is not automatically entirely nonbinding; its effect depends on the actual wording and law. An option to obtain rights is also distinct from an exercised option or an existing license grant.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

development milestone

A defined development achievement linked to review or agreed consequences.

IN USE specify development milestones

upfront payment

Consideration due at an initial agreed point in a transaction.

IN USE distinguish an upfront payment

royalty

A payment calculated on an agreed basis for use of licensed rights.

IN USE define the royalty basis

milestone consideration

Payment or value contingent on a specified contractual achievement.

IN USE clarify milestone consideration

cost sharing

Allocation of specified expenses between collaborating parties.

IN USE agree cost sharing

joint steering committee

A joint body with defined oversight or coordination responsibilities.

IN USE establish a joint steering committee

governance

The agreed structures and rules for overseeing a collaboration.

IN USE define collaboration governance

decision right

Authority to make a specified decision.

IN USE allocate decision rights

termination right

A contractual ability to end an arrangement under defined conditions.

IN USE review termination rights

rights reversion

Return or cessation of specified granted rights under agreed conditions.

IN USE define rights reversion

data room

An organized controlled-access collection of information for review.

IN USE maintain the data room

term sheet

A document summarizing proposed principal deal terms, with legal effect depending on its wording.

IN USE prepare a reviewed term sheet

Precision check

Scope includes more than the word exclusive: identify the rights, field, territory, duration, conditions, and retained uses. Governance also needs actual decision authority. A committee name alone does not establish who may approve a budget, change the program, or license an asset.

Phrases you can actually use

C Notice the language / 1 of 2

ACKNOWLEDGE THE INTEREST

"I understand that you want protection for the resources you would commit."

CHALLENGE THE BREADTH

"All platform applications is broader than the work currently proposed."

ASK ABOUT THE FIELD

"Which specific applications need exclusivity for the evaluation?"

PROTECT RETAINED USES

"We need to identify the rights and activities each party retains."

DEFINE THE FIRST STAGE

"The proposed evaluation covers T2 and T5 for six months."

SEPARATE DISCUSSION FROM GRANT

"I can discuss scope, but I cannot grant rights in this meeting."

CLARIFY DURATION

"What period would the proposed restriction cover?"

DISTINGUISH THE OPTION

"A future licensing option is not the same as a present commercial license."

08

Exclusive

Incomplete without the rights, activity, place, time, and conditions.

Option versus license

The ability to obtain rights later is not the same as already holding them.

Partner

A commercial relationship label does not define decision authority.

Phrases you can actually use

C Notice the language / 2 of 2

DEFINE THE TRIGGER

"Which evidence and conditions would permit an option to be exercised?"

CLARIFY FUNDING

"Which costs would each party bear during the evaluation?"

AVOID ASSUMED AGREEMENT

"Those terms are still open and subject to the required review."

CHECK THE DOCUMENT

"Identify which term-sheet provisions, if any, are intended to bind the parties."

CONNECT COMMITMENT

"Any exclusivity proposal needs appropriately defined obligations and review."

ASK ABOUT GOVERNANCE

"Who would have authority over scope, budget, and continuation decisions?"

LIMIT THE INFORMATION

"Provide data-room access within the agreed confidentiality and use conditions."

CLOSE THE MEETING

"We will prepare a narrower proposal without representing that rights are already granted."

08

Worldwide

Geographic breadth does not specify the field of use.

Milestone payment

The trigger comes from actual agreed terms, not any favorable laboratory result.

Agreed to discuss

A process commitment should not be recorded as a completed rights grant.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which counterproposal fits Nina's authority?

- A** Assign every future invention without review.
- B** Treat the request as an executed license.
- C** Prepare a reviewed proposal limited to the two-target, six-month evaluation and explicitly unresolved terms.
- D** Grant worldwide exclusive rights to all platform applications immediately.

2. Why distinguish an option from a license?

- A** An option removes the need to define any conditions.
- B** An option concerns a defined opportunity to obtain rights, while a license grants specified permission.
- C** Every option already grants all commercial rights.
- D** A license is always merely a meeting invitation.

3. A draft names a joint steering committee but gives it no powers. Which question addresses that gap?

- A** Which targets are included in the six-month evaluation?
- B** Which territories would the proposed exclusive rights cover?
- C** Which material will be placed in the controlled data room?
- D** Who may approve scope changes, budgets, and continuation decisions?

4. What should the meeting record say?

- A** The unapproved budget has already been paid.
- B** The parties will prepare a narrower proposal; no rights grant is approved.
- C** Worldwide exclusivity is effective because Tomas requested it.
- D** Every term-sheet provision is automatically nonbinding.

Answers and explanations: [page 99](#).

Define the collaboration before granting exclusivity

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

cost sharing / option agreement / exclusivity scope / joint steering committee / field restriction / term sheet / reserved rights / diligence obligation / rights reversion / option exercise

08

- 01 **Tomas:** Before we commit resources, we would like worldwide exclusive rights across the platform. We do not want to fund the work while a competitor receives the same opportunity.
- 02 **Nina:** I understand the concern, but the ¹ _____ you describe is wider than the proposed six-month evaluation of T2 and T5. Let us identify the activity you need protected.
- 03 **Tomas:** Our immediate interest is those two targets. We also see possible applications beyond them, and we would prefer not to renegotiate every time the opportunity develops.
- 04 **Nina:** A defined ² _____ could keep the first stage aligned with the actual work. Broader applications require their own review; I cannot grant platform-wide rights in this meeting.
- 05 **Tomas:** Could we start with evaluation rights and secure a route to a later commercial license if the results meet agreed criteria?
- 06 **Nina:** We can discuss an ³ _____ as a proposal. It would need clear scope, conditions, and approval; a possible future option is not already a commercial license.
- 07 **Tomas:** We would need to know what evidence qualifies and when we must decide. Otherwise the option could be too uncertain to support a meaningful investment.
- 08 **Nina:** Define the ⁴ _____ requirements alongside the evidence criteria. We should also distinguish scientific success from the separate commercial and legal conditions for obtaining rights.
- 09 **Tomas:** What happens to your own research during the evaluation? We do not necessarily need to stop unrelated platform work, but we need to understand competing activity.
- 10 **Nina:** Identify the ⁵ _____ and proposed restrictions explicitly. Neither side should infer a platform-wide freeze from a limited evaluation, nor assume an exception that the terms do not state.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 73.

- 11 **Tomas:** We can bring specialist capability and funding for the evaluation, but the division of costs has not been discussed. That needs to be part of the same proposal.
- 12 **Nina:** Yes, specify ⁶ _____ for the actual work. A broad expression of interest should not conceal who is paying for which activities or approving additional spending.
- 13 **Tomas:** If we seek some exclusivity, you will want assurance that the work moves forward. We should discuss what commitments would accompany that protection.
- 14 **Nina:** Appropriate ⁷ _____ terms can address that, subject to review. The proposal should state the expected progress and consequences rather than rely on a general promise to try.
- 15 **Tomas:** Could a joint committee handle routine technical decisions? I do not want a small adjustment to require another executive negotiation every time.
- 16 **Nina:** A ⁸ _____ can help if its authority is defined. We still need to assign decisions about scope, budget, continuation, and matters reserved for authorized signatories.
- 17 **Tomas:** We should also describe what happens if the evaluation ends without a later license. Materials, data access, and any temporary restrictions need a clear position.
- 18 **Nina:** Address ⁹ _____ and the relevant end-of-project obligations in the reviewed proposal. Do not assume that a temporary arrangement continues indefinitely because nobody discussed its end.
- 19 **Tomas:** I will tell our team we are preparing a two-target proposal, not announcing worldwide exclusivity. Send me the open points so we can assign reviewers on our side.
- 20 **Nina:** Exactly. We will prepare a ¹⁰ _____ covering the two-target evaluation and open issues, with its intended legal effect reviewed. No license, option, or exclusivity grant is approved by this discussion alone.

08

ANSWERS / EXPLANATIONS: PAGE 100

1. exclusivity scope | 2. field restriction | 3. option agreement | 4. option exercise | 5. reserved rights | 6. cost sharing | 7. diligence obligation | 8. joint steering committee | 9. rights reversion | 10. term sheet

An option is not an exercised grant

F Say it

Two partners discuss a possible option to negotiate a future license after a defined evaluation. No option is signed and no license is granted. A meeting note says commercial rights acquired.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

status / granted / signed / discussion

08

Partner: "The possible option is still under ¹ _____."

Lead: "No option has been ² _____."

Partner: "No commercial license has been ³ _____."

Lead: "The meeting note must preserve that ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 100

1. discussion | 2. signed | 3. granted | 4. status

Rehearse with a fixed script

1. Read the partnering exchange. Preserve two targets, six months, and Nina's authority to discuss but not grant rights.
2. Swap roles. Separate a possible option, its future exercise, and a current license; keep the terms unapproved.
3. Read the corrected meeting-note transfer. No option signed and no commercial license granted are the supplied statuses.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A familiar cell-line label is not an identity check

E More workplace conversations / 1-10

Discuss cell-bank identity and contamination status without overstating what a test establishes.

A research team wants to compare a newly received human cell stock with its existing working stock. Labels look similar, but the identity records and passage history have not been reconciled. A negative mycoplasma result is available only for the existing stock. This discussion concerns research readiness, not a laboratory protocol.

PEOPLE / Inez: Cell-platform scientist | Caleb: Project scientist

Use each bank entry once, without changing its form. Answers follow turn 20.

09

WORD + PHRASE BANK

mycoplasma testing / cross-contamination / working cell bank / cell-line authentication / passage history / provenance

- 01 **Caleb:** The new stock has the same cell-line name as ours. Can we combine its results with the current study?
- 02 **Inez:** Not on the label alone. We need the ¹ _____ records and a clear link to the material we actually received.
- 03 **Caleb:** Our current stock tested negative for mycoplasma. I assumed that covered the comparison.
- 04 **Inez:** That result concerns the tested stock and sampling point. It does not establish the new stock's identity or contamination status.
- 05 **Caleb:** Then identity testing and ² _____ answer different questions, even when both appear in the quality package.
- 06 **Inez:** Exactly. We need the relevant results, methods, dates, and material identifiers, with their limits intact.
- 07 **Caleb:** The old stock's label also has a passage number. The incoming paperwork lists a different number without explaining the counting convention.
- 08 **Inez:** Reconcile the ³ _____. Do not assume two identical numbers describe equivalent histories or that different numbers necessarily prove a problem.
- 09 **Caleb:** The supplier refers to a master bank, while our experiments use a working bank. Can you explain the relationship in the project summary?
- 10 **Inez:** The ⁴ _____ should have a traceable relationship to the source bank under our system.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 76.

- 11 **Caleb:** Could normal-looking cells be enough reassurance while we wait for the records?
- 12 **Inez:** No. Appearance may be an observation, but it does not rule out misidentification or
5 _____.
- 13 **Caleb:** We have not used the incoming stock yet. Should I tell the team the comparison is awaiting the readiness review?
- 14 **Inez:** Yes. Follow the site's handling and status controls, and keep the two stocks and their records distinct while the questions are resolved.
- 15 **Caleb:** I will request the missing information rather than rewrite the incoming label to match our familiar name.
- 16 **Inez:** Good. Preserve the ⁶ _____ and record any authorized correction with its basis.
- 17 **Caleb:** Once the records are reconciled, we still need to decide whether pooling the datasets answers the study question.
- 18 **Inez:** Correct. Confirmed identity does not by itself establish that all experimental conditions are comparable.
- 19 **Caleb:** I will separate the material-readiness question from the data-combination decision in the update.
- 20 **Inez:** That gives the team two clear checks instead of one reassuring but unsupported statement that the stocks are the same.

09

ANSWERS / EXPLANATIONS: PAGE 101

1. cell-line authentication | 2. mycoplasma testing | 3. passage history | 4. working cell bank | 5. cross-contamination | 6. provenance

A result from the old stock is attached to the new one

F Say it in a new situation

A spreadsheet links an existing stock's contamination-test result to a newly received stock. The new stock has not been assessed.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

questions / stock / history / unestablished

09

Scientist: This result belongs to the previously tested ¹ _____.

Reviewer: Restore the correct link and retain the correction ² _____.

Scientist: Mark the new stock's test status as ³ _____.

Reviewer: Keep identity and contamination as separate review ⁴ _____.

ANSWERS / EXPLANATIONS: PAGE 101

1. stock | 2. history | 3. unestablished | 4. questions

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

The sequencing batches also separate every study group

E More workplace conversations / 1-10

Explain confounding and challenge a confident biological interpretation of a batch-aligned result.

A fictional RNA-sequencing study placed every case sample in batch A and every control sample in batch B. The expression plot separates the groups. No batch-independent confirmation exists. The analyst and study scientist are discussing the interpretation and a possible follow-up design, not a validated diagnostic test.

PEOPLE / Ruth: Bioinformatics analyst | Dev: Study scientist

Use each bank entry once, without changing its form. Answers follow turn 20.

10

WORD + PHRASE BANK

confounding / technical replicates / balanced design / Biological replicates / batch effect / sample metadata

- 01 **Dev:** The expression plot separates cases and controls cleanly. Can we call this a disease signature in the report?
- 02 **Ruth:** Not yet. The groups also separate by processing batch, so a ¹ _____ could contribute to the pattern.
- 03 **Dev:** Every case was in A and every control in B. I thought the analysis could simply adjust for that afterward.
- 04 **Ruth:** Here we have complete ² _____ between group and batch. An adjustment cannot create the missing independent comparison.
- 05 **Dev:** Would sequencing each sample again count as more independent evidence for the disease difference?
- 06 **Ruth:** That would add ³ _____, depending on what is repeated. It would not create new independent biological samples by itself.
- 07 **Dev:** Then we need to distinguish the number of samples from the number of times each was measured.
- 08 **Ruth:** Yes. ⁴ _____ provide a different kind of information. The sample structure must stay visible in the analysis.
- 09 **Dev:** The plot is still useful for spotting the problem, even if we cannot interpret the separation as purely biological.
- 10 **Ruth:** Exactly. Keep the plot, annotate the batch assignment, and explain the design limitation rather than hide the uncomfortable grouping.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 79.

- 11 **Dev:** Could we remove genes that seem batch-sensitive until the case-control separation looks more convincing?
- 12 **Ruth:** Not as an outcome-driven cleanup. Filtering needs a defensible basis, and it cannot establish which part of a perfectly confounded pattern is biological.
- 13 **Dev:** For follow-up, I want the study team and analysis team to agree the assignments before sample processing begins.
- 14 **Ruth:** We should review a ⁵ _____ and the relevant constraints before generating more data.
- 15 **Dev:** We will also need enough metadata to reconstruct how every sample moved through processing.
- 16 **Ruth:** Correct. The ⁶ _____ must link reliably to the expression data, including any changes or exclusions.
- 17 **Dev:** What can we state now without implying the study is worthless?
- 18 **Ruth:** That the observed separation aligns with both disease group and batch, so this dataset alone cannot separate those explanations.
- 19 **Dev:** I will revise the headline and request review of the follow-up design. We will not present this plot as a validated disease classifier.
- 20 **Ruth:** Good. The limitation identifies what the next study needs to resolve; it does not require pretending we observed no pattern at all.

10

ANSWERS / EXPLANATIONS: PAGE 102

1. batch effect | 2. confounding | 3. technical replicates | 4. Biological replicates | 5. balanced design | 6. sample metadata

Repeated readings are counted as new donors

F Say it in a new situation

Samples from five donors are each measured twice. A summary reports ten independent donors. No additional donors were recruited.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

replicates / analysis / readings / metadata

10

Scientist: We have five donors, not ten independent biological ¹ _____.

Analyst: The extra measurements provide technical repeat ² _____.

Scientist: Keep the donor-to-measurement links in the ³ _____.

Analyst: Correct the sample-size claim before interpreting the ⁴ _____.

ANSWERS / EXPLANATIONS: PAGE 102

1. replicates | 2. readings | 3. metadata | 4. analysis

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

The dataset is coded, but the proposed use has changed

E More workplace conversations / 1-10

Clarify research-use permissions and data-sharing boundaries before a new collaboration.

A team has authorized access to coded human research data for a named study. A partner proposes a different analysis and requests a copy. The new purpose, partner access, and applicable consent limitations have not been reviewed. No transfer has occurred.

PEOPLE / Farah: Translational scientist | Ben: Research-governance lead

Use each bank entry once, without changing its form. Answers follow turn 20.

WORD + PHRASE BANK

withdrawal request / data-use limitations / data-use agreement / secondary-use / consent scope / Coded data

11

- 01 **Farah:** The partner can run the new analysis next week. Since the files have study codes rather than names, may I send a copy?
- 02 **Ben:** Not yet. ¹ _____ are not automatically unrestricted. What purpose and recipients does our current authorization cover?
- 03 **Farah:** It names our existing study. The partner wants to investigate a different outcome using the same records.
- 04 **Ben:** Then we need a ² _____ review before treating the proposal as covered.
- 05 **Farah:** The scientific question is related. Does that make it close enough to proceed while the paperwork catches up?
- 06 **Ben:** Related scientifically does not settle the permission. Check the ³ _____ and the applicable institutional and repository conditions.
- 07 **Farah:** Some records may have different restrictions. I had planned to send one combined file for convenience.
- 08 **Ben:** Map the ⁴ _____ before combining or transferring anything. We cannot erase differences by exporting them together.
- 09 **Farah:** The partner has a secure system and experienced analysts. Would security approval be sufficient?
- 10 **Ben:** Security is necessary, but it does not authorize the purpose or recipients. We also need the appropriate ⁵ _____ and access decisions.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 82.

- 11 **Farah:** Could I give them my login so they can inspect the fields without receiving a separate copy?
- 12 **Ben:** No. Use the approved access process with named authorized users. Sharing credentials would bypass the controls we are trying to verify.
- 13 **Farah:** I will prepare the proposed question, required fields, analysts, and planned outputs for review.
- 14 **Ben:** Include whether any new linkage is proposed and how outputs will be checked. We need the actual workflow, not just the project's title.
- 15 **Farah:** There is also a recent participant request concerning further use. I do not know how it affects copies or completed analyses.
- 16 **Ben:** Route that ⁶ _____ through the responsible team. Do not promise universal deletion or assume it has no effect.
- 17 **Farah:** For now, the partner gets no files or credentials. I can explain that the new analysis needs its own review.
- 18 **Ben:** Yes. Preserve the current access boundaries while the reviewers establish which records and uses, if any, are permitted.
- 19 **Farah:** Once the decision arrives, I will confirm the exact approved dataset and recipient list before arranging the transfer.
- 20 **Ben:** And retain the authorization with the transfer record. Scientific enthusiasm is valuable, but the permitted scope must remain traceable.

11

ANSWERS / EXPLANATIONS: PAGE 103

1. Coded data | 2. secondary-use | 3. consent scope | 4. data-use limitations | 5. data-use agreement | 6. withdrawal request

A new collaborator requests an existing login

F Say it in a new situation

A collaborator is not yet an approved user of a controlled-access research dataset and asks to borrow a team member's login.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

scope / credentials / process / use

Scientist: Existing access does not authorize sharing our ¹ _____.

Governance lead: Submit the collaborator through the named-user access
² _____.

Scientist: We must also confirm that the proposed question is a permitted
³ _____.

Governance lead: Keep the data within the current authorized ⁴ _____
meanwhile.

11

ANSWERS / EXPLANATIONS: PAGE 103

1. credentials | 2. process | 3. use | 4. scope

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

01 Briefing and language checks

Platform Technology and Scientific Thesis

A Briefing checks / page 4

1. Which evidence is supplied?

D. A 40% average reporter-signal reduction in three independent preparations of one cell line

The brief supplies a reporter result in one cell line, not cross-disease or clinical evidence.

2. Which claim exceeds that evidence?

D. Proven activity across three diseases

The additional disease systems have not been tested in the supplied case.

3. What is the immediate decision request?

D. Funding a bounded follow-up study

The stated request concerns a limited next research study, not a clinical or rights decision.

D Language checks / page 9

1. Which slide headline is best supported?

B. Reporter signal reduced by an average 40% in the tested cell line; broader applications remain untested.

The supported headline retains the observed result and its actual experimental boundary.

2. One preparation is read three times with the same method. How should those reads be described?

D. Three technical measurements of one biological preparation

Repeated reads of the same material are technical replication. They do not create independently prepared biological units, change the measurement approach, or establish new disease applications.

3. Why request an orthogonal functional readout?

C. To examine the biological question through a different measurement approach

A different approach can test whether the original signal reflects the intended biology.

4. Which funding request matches the brief?

D. Support a bounded follow-up addressing functional confirmation and broader relevance.

The brief asks for a next research study that addresses the stated evidence gaps.

01 Conversation answers

Platform Technology and Scientific Thesis

E Dialogue gaps

Pages 10-11

1. reporter signal. Turn 2. The reporter signal is the measured experimental output, not a direct demonstration of clinical benefit.

2. scientific thesis. Turn 4. The scientific thesis connects the observation to a research argument without treating every hypothesis as established.

3. biological replicate. Turn 6. A biological replicate is an independent preparation within the tested system, not a different disease application.

4. proof of mechanism. Turn 8. Proof of mechanism requires evidence for the proposed biological explanation beyond the supplied reporter observation.

5. orthogonal assay. Turn 10. An orthogonal assay uses a different measurement approach to examine the underlying biological question.

6. assay interference. Turn 12. Assay interference can alter the measured signal without establishing the intended biological effect.

7. translational hypothesis. Turn 14. A translational hypothesis proposes a link to another development setting that remains to be tested.

8. evidence gap. Turn 16. The evidence gap is the missing functional and broader-system support needed for the larger claim.

9. decision gate. Turn 18. A decision gate defines when the follow-up evidence will be reviewed against agreed continuation criteria.

10. generalizability. Turn 20. Generalizability concerns application beyond the tested setting, which the supplied evidence has not yet established.

F Transfer conversation

Page 12

1. model

Scientist: "The signal was observed in one model."

The supplied result belongs to one tested cell model only.

2. untested

Partner: "The other two are still untested."

The briefing states that no testing has occurred in those additional systems.

3. hypothesis

Scientist: "Broader application remains a hypothesis."

The proposed extension is a testable idea rather than an established result.

4. gap

Partner: "The next study should address that evidence gap."

The missing data concern whether the finding extends beyond the tested setting.

02 Briefing and language checks

Assay Development and Reproducibility

A Briefing checks / page 13

1. What is consistent within each run?

B. The stated coefficient of variation is 5%.

The briefing gives a five-percent within-run coefficient of variation on each day.

2. What changes across days?

A. The mean signal: 80, 100, and 125 units

The day means differ even though the within-run variation is stated as the same.

3. What cannot yet be concluded?

A. A confirmed cause or full validation status

Neither a cause nor acceptance criteria sufficient for a validation decision are provided.

D Language checks / page 18

1. Which statement fits the data?

B. Each run has 5% CV, while day means differ from 80 to 125 units.

The correct statement keeps within-run spread separate from the observed between-day differences.

2. Relative to day two's mean of 100, how much higher is day three's mean of 125?

A. 25%

The increase is twenty-five divided by the reference value one hundred, giving twenty-five percent.

3. Which follow-up best supports investigation?

A. Compare traceable run conditions and raw data before assigning a cause.

Comparing records and raw data can test explanations without prematurely choosing one.

4. A colleague rescales each day to a mean of 100 and calls the assay reproducible. What remains unresolved?

D. Whether the adjustment has a justified basis and the original between-day variation is understood

Equalizing the plotted means is the adjustment's result, not independent evidence of performance. The original shift and the scientific basis for normalization still require examination.

02 Conversation answers

Assay Development and Reproducibility

E Dialogue gaps

Pages 19-20

- 1. within-run variation.** Turn 2. Within-run variation describes spread inside each run, not whether the averages match across different days.
- 2. coefficient of variation.** Turn 4. The coefficient of variation expresses relative spread as standard deviation divided by the mean.
- 3. run metadata.** Turn 6. Run metadata identifies conditions such as operator, reagent lot, and calibration that can support investigation.
- 4. root-cause investigation.** Turn 8. A root-cause investigation tests supported explanations rather than assigning responsibility from an incomplete pattern.
- 5. intermediate precision.** Turn 10. Intermediate precision covers relevant variation within one laboratory, including between-day conditions.
- 6. mean signal.** Turn 12. Mean signal is the average measured output being compared between day two and day three.
- 7. acceptance criterion.** Turn 14. An acceptance criterion supplies the predefined requirement needed for a formal performance judgment.
- 8. normalization.** Turn 16. Normalization requires a defined and justified reference basis rather than an adjustment chosen only to conceal disagreement.
- 9. calibration.** Turn 18. Calibration records describe the relationship to reference values and may help explain measurement differences.
- 10. precision.** Turn 20. Precision concerns agreement among repeated measurements under specified conditions and must be qualified by those conditions.

F Transfer conversation

Page 21

1. thirty

Analyst: "The mean rose by thirty units."

Subtracting one hundred twenty from one hundred fifty gives thirty units.

2. 25%

Reviewer: "Relative to 120, that is a 25% increase."

Thirty divided by the reference value one hundred twenty equals twenty-five percent.

3. spread

Analyst: "The four-percent CV describes within-run spread."

The coefficient of variation concerns relative spread, not the change between means.

4. criteria

Reviewer: "A formal acceptance decision still needs the applicable criteria."

The briefing supplies no acceptance criteria against which to make that decision.

03 Briefing and language checks

Biomarkers and Translational Strategy

03

A Briefing checks / page 22

1. Which claim is supported by the supplied study?

B. The test met predefined analytical criteria over the assessed range and plasma conditions.

The supplied study concerns measurement performance in a defined analytical setting.

2. What evidence is absent?

B. A study establishing prediction of response to B17

The briefing expressly says response prediction has not been established.

3. Which proposed use requires additional support?

C. Selecting participants on an asserted predictive role

Patient selection based on response prediction goes beyond the supplied analytical evidence.

D Language checks / page 27

1. Which description matches the supplied evidence?

C. The test met analytical criteria from 2 to 20 research units; prediction of B17 response is unestablished.

The correct description separates the completed analytical evaluation from the unestablished clinical relationship.

2. Why is a predictive label premature?

D. No study has established differing B17 response associated with the marker.

A predictive claim requires evidence about the intervention-related response, which the brief says is absent.

3. A colleague chooses 11 as an eligibility cutoff because it is the midpoint of 2 to 20. What evidence is missing?

D. Evidence supporting classification by that cutoff for the proposed B17 use

An in-range midpoint is an arithmetic property, not validation of a selection rule. The missing evidence concerns the intended response-related clinical use.

4. What would changing the sample from plasma require in the claim?

D. Appropriate evaluation of the changed matrix rather than automatic transfer of performance

Sample components can affect measurement, so the original conditions cannot be generalized automatically.

03 Conversation answers

Biomarkers and Translational Strategy

03

E Dialogue gaps

Pages 28-29

- 1. analytical validation.** Turn 2. Analytical validation concerns the test's measurement performance under its specified conditions and intended analytical use.
- 2. clinical validation.** Turn 4. Clinical validation would establish the relationship to the proposed clinical concept, here B17 response prediction.
- 3. context of use.** Turn 6. Context of use specifies the purpose and setting of the proposed biomarker application.
- 4. analyte.** Turn 8. The analyte is the substance measured, allowing a precise analytical claim without an unsupported predictive label.
- 5. decision threshold.** Turn 10. A decision threshold needs support for its classification purpose and is not supplied by the midpoint of an analytical range.
- 6. prognostic biomarker.** Turn 12. A prognostic biomarker concerns clinical course and does not automatically identify differential effects of a treatment.
- 7. predictive biomarker.** Turn 14. A predictive biomarker identifies different likelihoods of effects from a particular intervention, requiring the relevant evidence.
- 8. matrix effect.** Turn 16. Matrix effect describes how other sample components can influence measurement, limiting automatic transfer between sample types.
- 9. Biomarker qualification.** Turn 18. Biomarker qualification is a formal, scoped acceptance process, not a synonym for an internal analytical result.
- 10. clinical utility.** Turn 20. Clinical utility asks whether applying the test is useful in the specific care decision, beyond measurement alone.

F Transfer conversation

Page 30

1. range

Scientist: "Five to fifty is the assessed measurement range."

The supplied interval describes where measurement performance was evaluated.

2. threshold

Colleague: "Twenty-five is a proposed decision threshold."

The proposed cutoff concerns an eligibility decision, not merely measurement capability.

3. unstudied

Scientist: "Its relationship to response is unstudied."

The briefing explicitly says no response relationship has been studied.

4. selection

Colleague: "The analytical result does not authorize participant selection."

Measurement performance alone does not support the proposed clinical selection rule.

04 Briefing and language checks

Preclinical Data Packages

04

A Briefing checks / page 31

1. What is the reported comparison?

C. A 20% lower mean model-endpoint value in 12 treated animals versus 12 controls at one exposure

The brief describes an animal-model mean endpoint comparison, not a human outcome.

2. What is known about blinding?

C. It is not documented in the supplied summary.

The summary's omission leaves blinding status unestablished rather than proving either condition.

3. Which conclusion is unsupported?

D. Human benefit has been proved.

No human outcome data are supplied, so the animal result cannot establish human benefit.

D Language checks / page 36

1. Which sentence best represents the result?

A. At one exposure, the treated animal group had a 20% lower mean model endpoint than controls; human benefit is unestablished.

The supported sentence preserves the model, comparison, exposure limit, and absence of human evidence.

2. What does the missing blinding information mean?

A. The review needs clarification from the study record.

An omitted design detail needs verification rather than an assumption about how the study was conducted.

3. Which addition would directly clarify uncertainty around the reported 20% model-endpoint difference?

D. The underlying group results and a justified uncertainty analysis

Group sizes and randomization matter but do not alone quantify uncertainty in the observed difference. The results and appropriate analysis address that question; a target population does not replace them.

4. Which distinction is accurate?

A. Administered dose and measured exposure are related but not identical concepts.

Dose describes administration, while exposure describes the substance reaching the system over time.

04 Conversation answers

Preclinical Data Packages

E Dialogue gaps

Pages 37-38

1. efficacy endpoint. Turn 2. The efficacy endpoint is the defined model measurement, not a directly measured human clinical outcome.

2. model relevance. Turn 4. Model relevance concerns how the experimental system relates to the development question without equating it with people.

3. Randomization. Turn 6. Randomization concerns chance-based allocation and does not establish whether assessors knew group assignments.

4. study report. Turn 8. The full study report provides methods and results needed to verify details omitted from a short summary.

5. uncertainty interval. Turn 10. An uncertainty interval expresses the estimate's uncertainty under the relevant analytical method.

6. exposure-response relationship. Turn 12. An exposure-response relationship requires evidence about how effects vary with exposure, not one tested condition alone.

7. pharmacokinetics. Turn 14. Pharmacokinetics concerns the substance's disposition over time and helps distinguish dose from resulting exposure.

8. toxicology. Turn 16. Toxicology concerns harmful effects and cannot be replaced by a favorable efficacy-endpoint result.

9. safety margin. Turn 18. A safety margin needs a justified comparison of relevant levels, which the supplied facts do not provide.

10. translational uncertainty. Turn 20. Translational uncertainty limits the extension from the tested model to another biological or clinical setting.

F Transfer conversation

Page 39

1. allocation

Scientist: "Random allocation is documented."

The supplied summary explicitly describes random assignment to groups.

2. unstated

Reviewer: "Assessor blinding remains unstated."

The summary gives no information establishing the assessors' blinding status.

3. record

Scientist: "We need the relevant study record."

The underlying study documentation can clarify the omitted design detail.

4. verification

Reviewer: "Neither proposed label is supported without verification."

Both fully blinded and definitely unblinded exceed the currently supplied evidence.

04

05 Briefing and language checks

CMC and Scale-Up in Biotech

A Briefing checks / page 40

1. What was demonstrated?

C. Harvest titer of 0.8 g/L in two 1-liter runs

The evidence is limited to harvest titer in the two specified small runs.

2. What does 0.8 g/L multiplied by 200 liters give under the stated assumption?

C. A hypothetical 160 grams at harvest before downstream losses

The arithmetic estimates harvest quantity if the same titer holds; recovery and quality are separate.

3. Which evidence remains missing?

D. Larger-scale performance, downstream recovery, and relevant quality results

The briefing expressly leaves those larger-scale and finished-product conditions unestablished.

D Language checks / page 45

1. Which forecast is appropriately qualified?

B. If 0.8 g/L holds at 200 liters, harvest quantity would be 160 grams before recovery and quality evaluation.

The qualified forecast states its titer assumption, process stage, and unresolved finished-product conditions.

2. If a separate hypothetical process retains 70% of 160 grams, how much remains after that step?

A. 112 grams

Multiplying one hundred sixty by zero point seven gives one hundred twelve grams.

3. Which item is a quality question rather than simply a quantity calculation?

B. Whether relevant potency and impurity attributes remain appropriate

Potency and impurity attributes concern product quality, not only arithmetic output.

4. Both one-liter runs have the same titer. Which proposed summary overstates the demonstrated scope?

C. Matching titer establishes larger-scale quality comparability.

Equal small-run concentration is not evidence for all relevant larger-scale quality attributes. The other statements retain the observed scale or explicitly identify the open assumptions.

05 Conversation answers

CMC and Scale-Up in Biotech

E Dialogue gaps

Pages 46-47

- 1. titer.** Turn 2. Titer is concentration at the specified harvest stage, not a quantity of finished released material.
- 2. scale-up.** Turn 4. Scale-up changes process size and requires relevant performance evidence rather than automatic extrapolation from small runs.
- 3. recovery.** Turn 6. Recovery describes the proportion retained through specified downstream stages and affects the final quantity.
- 4. critical quality attribute.** Turn 8. Critical quality attributes concern properties necessary for intended quality, beyond the amount produced.
- 5. Potency.** Turn 10. Potency measures relevant biological activity and is not established solely by product concentration.
- 6. mass transfer.** Turn 12. Mass transfer concerns movement within the system and may behave differently when the process scale changes.
- 7. scale-down model.** Turn 14. A scale-down model can represent selected larger-process behavior when its relevance is appropriately established.
- 8. change control.** Turn 16. Change control provides the formal review and traceability needed when process changes are proposed or implemented.
- 9. comparability.** Turn 18. Comparability requires assessment of relevant product consequences of a change, not agreement in a single output alone.
- 10. CMC.** Turn 20. CMC is the chemistry, manufacturing, and controls package connecting process performance with product quality.

F Transfer conversation

Page 48

1. fifty

Scientist: "The harvest contains fifty grams."

One hundred liters multiplied by zero point five grams per liter equals fifty grams.

2. forty

Reviewer: "At eighty-percent recovery, forty grams remain."

Eighty percent of fifty grams equals forty grams after the specified step.

3. release

Scientist: "That is a quantity calculation, not a release decision."

The briefing gives no quality review or authorization releasing the material.

4. separate

Reviewer: "Keep the relevant product-quality evidence separate."

The amount retained does not by itself establish the relevant quality attributes.

05

06 Briefing and language checks

IP, Freedom to Operate, and Collaboration

A Briefing checks / page 49

1. Which agreement is signed?

D. The confidentiality agreement

The brief states that confidentiality is agreed while the material-transfer terms remain a draft.

2. Which issue remains unresolved?

B. Permitted use and publication review, among other material-transfer terms

The briefing explicitly identifies the listed terms as unresolved.

3. What does ownership of a related patent establish about third-party rights here?

B. It does not by itself establish freedom to operate.

A party's own patent does not automatically resolve rights held by others.

D Language checks / page 54

1. Which response best fits the shipment question?

B. The MTA terms remain unresolved, so seek the required review and authorization before treating shipment as approved.

The brief establishes only a signed confidentiality agreement and unfinished transfer terms.

2. Which review question concerns permitted use rather than a different transfer term?

C. Which named bench assessments may the recipient perform with the protein?

Permitted use defines the recipient's authorized activities. The other questions concern onward transfer, publication review, and ownership; each is important but distinct.

3. Why does a related owned patent not settle FTO?

D. Other parties may hold relevant rights affecting the proposed acts.

Rights held by others can remain relevant despite ownership of a separate patent.

4. Which statement preserves the publication issue?

B. The publication-review process and timing remain to be agreed.

The supplied draft has unresolved publication terms, so neither unrestricted release nor permanent prohibition is established.

06 Conversation answers

IP, Freedom to Operate, and Collaboration

E Dialogue gaps

Pages 55-56

1. material transfer agreement. Turn 2. The material transfer agreement governs the proposed tangible-material transfer and remains unfinished in this case.

2. nondisclosure agreement. Turn 4. A nondisclosure agreement addresses specified confidential information and does not automatically settle material-use permission.

3. permitted use. Turn 6. Permitted use identifies which activities the recipient is authorized to perform under the agreement.

4. background IP. Turn 8. Background IP identifies pre-existing intellectual property brought into the collaboration under the agreed definitions.

5. Inventorship. Turn 10. Inventorship identifies qualifying contributors to a claimed invention and is distinct from ownership of the rights.

6. publication review. Turn 12. Publication review is the agreed process for examining proposed disclosure, with its purpose and timing specified.

7. know-how. Turn 14. Know-how concerns practical technical information whose permitted use may differ from possession of a material.

8. freedom to operate. Turn 16. Freedom to operate concerns the proposed acts and relevant rights of others, not merely ownership of one's own patent.

9. third-party rights. Turn 18. Third-party rights are held outside these two partners and can remain relevant to the proposed activities.

10. governing agreement. Turn 20. The governing agreement supplies the actual applicable terms for the authorized collaboration rather than an informal assumption.

F Transfer conversation

Page 57

1. evaluation

Scientist: "The stated purpose is evaluation."

The supplied permission concerns evaluating the confidential method summary.

2. permitted

Partner: "Commercial use is not established as permitted."

The brief expressly says commercial-use permission has not been agreed.

3. unresolved

Scientist: "Onward disclosure also remains unresolved."

No onward-disclosure permission is supplied, so it cannot be assumed.

4. terms

Partner: "We need the actual governing terms."

The relevant permissions depend on the applicable agreement rather than access alone.

07 Briefing and language checks

Investor and Board Updates

A Briefing checks / page 58

1. Why is the funding criterion not yet met?

A. Only two of the required three independent functional runs are complete.

The agreed criterion requires three consistent functional runs, and the third remains pending.

2. What is the simple base-case runway at unchanged net burn?

B. Ten months

Twelve million divided by one point two million per month equals ten months.

3. How should the \$0.3 million proposal be described?

C. Additional proposed funding, not yet approved or included in the base case

The brief explicitly leaves the proposed budget unapproved and outside the current base case.

D Language checks / page 63

1. Which opening is most decision-ready?

A. Two functional runs are complete; the third required by the funding criterion is pending, and we request \$0.3 million for the defined follow-up.

The opening connects actual evidence, the unmet criterion, and the specific proposed decision.

2. What makes the ten-month runway conditional?

D. It assumes unchanged net burn and the stated available cash basis.

The arithmetic holds only under the cash and burn assumptions stated in the brief.

3. The board packet counts six completed experiments against the three-run gate. What is the reporting error?

C. It combines different evidence categories although the criterion specifies functional runs.

Four binding experiments plus two functional runs is six activities, but only two match the specified category. The third functional run is still pending; a total count does not satisfy the gate.

4. What should happen if the new budget is approved?

D. Update the cash model for the actual spending assumptions and record the decision.

An approved additional spend changes planning inputs without establishing the experimental outcome.

07 Conversation answers

Investor and Board Updates

E Dialogue gaps

Pages 64-65

- 1. milestone criterion.** Turn 2. The milestone criterion specifies three consistent independent functional runs rather than a total activity count.
- 2. evidence package.** Turn 4. The evidence package organizes the relevant results for the decision without treating unlike experiments as interchangeable.
- 3. scientific risk.** Turn 6. Scientific risk concerns whether the biological or technical premise will hold, which completed activity alone cannot eliminate.
- 4. board ask.** Turn 8. The board ask is the specific bounded funding decision being requested, distinct from reporting completed work.
- 5. cash runway.** Turn 10. Cash runway estimates the duration of available cash under the stated net-burn assumptions.
- 6. base case.** Turn 12. The base case is the current central planning scenario, which excludes the unapproved additional proposal here.
- 7. contingent funding.** Turn 14. Contingent funding depends on future conditions and is not equivalent to cash already received.
- 8. dependency.** Turn 16. A dependency identifies the evidence or task on which the next decision relies.
- 9. downside case.** Turn 18. A downside case shows the consequences of specified less-favorable assumptions without treating them as established outcomes.
- 10. decision record.** Turn 20. The decision record preserves what the board actually decided and any conditions attached to it.

F Transfer conversation

Page 66

1. six

Finance lead: "The simple base-case runway is six months."

Nine million divided by one point five million per month equals six months.

2. burn

Program lead: "That assumes an unchanged net burn."

The calculation holds only at the stated net cash-outflow rate.

3. excluded

Finance lead: "The new project cost is currently excluded."

The briefing states that the proposed cost is outside the base case.

4. model

Program lead: "Approval would require an updated cash model."

Adding approved spending requires the planning assumptions and resulting cash estimate to be updated.

08 Briefing and language checks

Partnering and Business Development

A Briefing checks / page 67

1. What is the proposed first-stage work?

C. A six-month nonclinical evaluation of T2 and T5

The stated initial proposal is limited to two targets and a six-month nonclinical evaluation.

2. What authority does Nina have?

A. Discuss a narrower proposal, not grant rights

The brief expressly limits Nina to discussion rather than granting rights.

3. What remains unapproved?

C. License, option, budget, and governance terms

The briefing identifies those substantive agreement terms as not approved.

D Language checks / page 72

1. Which counterproposal fits Nina's authority?

C. Prepare a reviewed proposal limited to the two-target, six-month evaluation and explicitly unresolved terms.

Nina may discuss and prepare a narrower proposal but is not authorized to grant rights.

2. Why distinguish an option from a license?

B. An option concerns a defined opportunity to obtain rights, while a license grants specified permission.

The two arrangements confer different positions, and the actual terms determine their scope and exercise.

3. A draft names a joint steering committee but gives it no powers. Which question addresses that gap?

D. Who may approve scope changes, budgets, and continuation decisions?

Governance requires actual decision authority and rules. Targets, territory, and data access are separate scope questions and do not establish what the committee may decide.

4. What should the meeting record say?

B. The parties will prepare a narrower proposal; no rights grant is approved.

The record must preserve discussion status and the absence of an authorized grant.

08 Conversation answers

Partnering and Business Development

E Dialogue gaps

Pages 73-74

- 1. exclusivity scope.** Turn 2. Exclusivity scope defines what is protected and cannot be inferred from the broad word exclusive alone.
- 2. field restriction.** Turn 4. A field restriction limits the proposed rights to specified applications rather than every possible platform use.
- 3. option agreement.** Turn 6. An option agreement could provide a defined opportunity to obtain rights later under specified conditions.
- 4. option exercise.** Turn 8. Option exercise is the use of the option in accordance with its specified timing and conditions.
- 5. reserved rights.** Turn 10. Reserved rights are the uses or permissions retained rather than included in the proposed grant.
- 6. cost sharing.** Turn 12. Cost sharing allocates the defined evaluation expenses and cannot be assumed from general interest in partnering.
- 7. diligence obligation.** Turn 14. A diligence obligation defines the agreed efforts or progress expected rather than leaving exclusivity without operational commitments.
- 8. joint steering committee.** Turn 16. A joint steering committee can coordinate the collaboration, but its authority must be specified in the agreement.
- 9. rights reversion.** Turn 18. Rights reversion concerns the return or cessation of specified rights under agreed end conditions.
- 10. term sheet.** Turn 20. A term sheet summarizes proposed principal terms whose binding effect must be explicitly reviewed.

F Transfer conversation

Page 75

1. discussion

Partner: "The possible option is still under discussion."

The briefing states that the option is being discussed and is not signed.

2. signed

Lead: "No option has been signed."

The supplied facts explicitly deny a signed option agreement.

3. granted

Partner: "No commercial license has been granted."

The briefing states that no license grant has occurred.

4. status

Lead: "The meeting note must preserve that status."

The note should reflect discussion rather than falsely report acquired commercial rights.

08

09 Conversation answers

A familiar cell-line label is not an identity check

E Extended conversation

Pages 76-77

1. cell-line authentication. Turn 2. Cell-line authentication establishes identity through appropriate evidence; a matching label alone is not an identity check.

2. mycoplasma testing. Turn 5. Mycoplasma testing checks for a particular contamination concern; it is not interchangeable with establishing the cell line's identity.

3. passage history. Turn 8. Passage history records the sequence of cell subculturing and relevant lineage information; inconsistent counting needs clarification.

4. working cell bank. Turn 10. A working cell bank supplies routine work from a defined cell-bank lineage; its relationship to the source bank must remain traceable.

5. cross-contamination. Turn 12. Cross-contamination introduces material from another source and may not be excluded by normal-looking morphology alone.

6. provenance. Turn 16. Provenance identifies the material's origin and history; relabeling without resolving the discrepancy would hide that information.

F Transfer exchange

Page 78

1. stock

Scientist: This result belongs to the previously tested stock.

The record identifies the old stock and does not assess the new material.

2. history

Reviewer: Restore the correct link and retain the correction history.

Traceable correction preserves how the wrong association was identified and fixed.

3. unestablished

Scientist: Mark the new stock's test status as unestablished.

No assessment of the new stock has been supplied, so its status cannot be inferred.

4. questions

Reviewer: Keep identity and contamination as separate review questions.

Material identity and a specific contamination result are distinct evidence questions.

Background reading: [ATCC: Cell Line Authentication Test Recommendations](#). The conversation is an original fictional teaching case.

10 Conversation answers

The sequencing batches also separate every study group

E Extended conversation

Pages 79-80

- 1. batch effect.** Turn 2. A batch effect is systematic variation associated with processing groups rather than necessarily the biology under study.
- 2. confounding.** Turn 4. Confounding means the biological group and processing batch vary together, preventing their separate effects from being identified from this design alone.
- 3. technical replicates.** Turn 6. Technical replicates repeat measurement of the same biological material; they do not create additional independent biological samples.
- 4. Biological replicates.** Turn 8. Biological replicates are independent biological samples that capture relevant biological variation rather than repeated readings of one sample.
- 5. balanced design.** Turn 14. A balanced design distributes relevant biological groups across processing conditions so those factors are not automatically identical.
- 6. sample metadata.** Turn 16. Sample metadata describe origin, grouping, processing, and other relevant characteristics needed to understand the analysis design.

F Transfer exchange

Page 81

1. replicates

Scientist: We have five donors, not ten independent biological replicates.

Repeating measurements of the same five donors does not add new independent donors.

2. readings

Analyst: The extra measurements provide technical repeat readings.

The repeated measurements add readings of existing biological material rather than new donor sources.

3. metadata

Scientist: Keep the donor-to-measurement links in the metadata.

Metadata must preserve which measurements came from the same donor or sample.

4. analysis

Analyst: Correct the sample-size claim before interpreting the analysis.

The analysis must reflect the real dependence and number of independent samples.

10

Background reading: [EMBL-EBI: Experimental Design for RNA-seq](#). The conversation is an original fictional teaching case.

11 Conversation answers

The dataset is coded, but the proposed use has changed

E Extended conversation

Pages 82-83

1. Coded data. Turn 2. Coded data use identifiers such as study codes; coding does not by itself make every use or transfer unrestricted.

2. secondary-use. Turn 4. Secondary-use review assesses a further research purpose beyond the initially authorized use, rather than assuming access covers every question.

3. consent scope. Turn 6. Consent scope defines the relevant uses participants agreed to and any limits that must be respected in the proposed research.

4. data-use limitations. Turn 8. Data-use limitations specify restrictions attached to the records or consent groups; combining files does not remove those restrictions.

5. data-use agreement. Turn 10. A data-use agreement records permitted handling and use between parties; it must align with the governing permissions rather than replace them.

6. withdrawal request. Turn 16. A withdrawal request requires review under the relevant consent, policy, and legal framework; its effects should not be invented or ignored.

F Transfer exchange

Page 84

1. credentials

Scientist: Existing access does not authorize sharing our credentials.

Credentials identify authorized access and must not be lent to bypass user approval.

2. process

Governance lead: Submit the collaborator through the named-user access process.

The collaborator needs the applicable review and authorization for their own access.

3. use

Scientist: We must also confirm that the proposed question is a permitted use.

User approval and permitted research purpose are related but separate requirements.

4. scope

Governance lead: Keep the data within the current authorized scope meanwhile.

The unresolved proposal does not expand the existing authorization while review is pending.

11

Background reading: [NIH dbGaP: Data Access Requests and Data-Use Limitations](#). The conversation is an original fictional teaching case.

Find the move you need

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

01 Platform Technology and Scientific Thesis

Lead with the observation: We observed a 40% average reduction in the reporter signal.

Separate signal from function: The reporter result does not yet establish the functional effect.

Describe the control: The percentage is relative to the control specified in this experiment.

Full phrase set: pages 7-8 | Dialogue: pages 10-11

02 Assay Development and Reproducibility

Separate the levels: The within-run precision is consistent, but the day means differ.

Limit the conclusion: That does not establish agreement across days.

Investigate conditions: Compare reagent lots, calibration, operator, and relevant run conditions.

Full phrase set: pages 16-17 | Dialogue: pages 19-20

03 Biomarkers and Translational Strategy

Ask for the scope: Validated for which measurement and intended use?

Preserve the conditions: That evidence applies to the tested plasma conditions.

Use the correct label: Describe an analytically evaluated test, not a clinically validated predictive biomarker.

Full phrase set: pages 25-26 | Dialogue: pages 28-29

04 Preclinical Data Packages

Name the model result: The treated group had a 20% lower mean model-endpoint value than controls.

Keep the species boundary: Human benefit has not been established by these data.

Avoid a significance claim: The supplied percentage alone does not establish statistical significance.

Full phrase set: pages 34-35 | Dialogue: pages 37-38

Keep the conversation productive

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

05 CMC and Scale-Up in Biotech

State the observed scale: Both results came from one-liter runs.

Expose the assumption: The larger process has not yet demonstrated that titer.

Ask about attributes: Which quality attributes need comparison after this change?

Full phrase set: pages 43-44 | Dialogue: pages 46-47

06 IP, Freedom to Operate, and Collaboration

Identify the signed document: The confidentiality agreement is signed; the material-transfer terms are not final.

Check onward transfer: May the recipient send the material to another laboratory?

Avoid an unlimited delay: Publication review needs defined terms, not an unstated permanent veto.

Full phrase set: pages 52-53 | Dialogue: pages 55-56

07 Investor and Board Updates

Lead with the decision: We are asking for a bounded follow-up budget, not declaring the milestone complete.

State the gap: The third functional run is pending.

State approval status: The additional \$0.3 million has not yet been approved.

Full phrase set: pages 61-62 | Dialogue: pages 64-65

08 Partnering and Business Development

Acknowledge the interest: I understand that you want protection for the resources you would commit.

Separate discussion from grant: I can discuss scope, but I cannot grant rights in this meeting.

Distinguish the option: A future licensing option is not the same as a present commercial license.

Full phrase set: pages 70-71 | Dialogue: pages 73-74

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

acceptance criterion	specify acceptance criteria	15
accuracy	evaluate analytical accuracy	14
aggregation	monitor product aggregation	41
analyte	identify the analyte	23
analytical validation	complete analytical validation	23
assay development	plan assay development	14
assay drift	monitor assay drift	15
assay interference	investigate assay interference	6
assignment	distinguish an assignment from a license	50
background IP	define background IP	50
base case	update the base case	60
batch record	review the batch record	42
between-day variation	investigate between-day variation	14
bias	investigate measurement bias	14
bioavailability	assess bioavailability	32
bioburden	monitor bioburden	42
biodistribution	characterize biodistribution	33
biological replicate	include biological replicates	5
biomarker	evaluate a biomarker	23
biomarker qualification	define the qualification scope	23
blinding	verify assessor blinding	32
board ask	state the board ask	59
budget variance	explain a budget variance	60
calibration	verify calibration records	15

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

cash runway estimate cash runway	60
cell line characterize the cell line	5
change control follow change control	42
clinical utility assess clinical utility	23
clinical validation plan clinical validation	23
CMC review the CMC package	41
coefficient of variation report the coefficient of variation	14
comparability establish a comparability strategy	42
confidentiality obligation comply with confidentiality obligations	50
context of use specify the context of use	23
contingent funding identify contingent funding	60
control condition specify the control condition	5
cost sharing agree cost sharing	69
counter-screen include a counter-screen	6
critical path identify the critical path	59
critical process parameter assess critical process parameters	41
critical quality attribute identify critical quality attributes	41
data provenance preserve data provenance	33
data room maintain the data room	69
decision gate set a decision gate	6
decision record maintain a decision record	60
decision right allocate decision rights	69
decision threshold justify a decision threshold	24
dependency make dependencies explicit	59

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

development milestone	specify development milestones	69
diagnostic sensitivity	estimate diagnostic sensitivity	24
diagnostic specificity	estimate diagnostic specificity	24
diligence obligation	negotiate diligence obligations	68
dilution	explain potential dilution	60
disease model	qualify the disease model	6
dose-response relationship	distinguish dose-response from exposure-response	33
downside case	test the downside case	60
downstream processing	evaluate downstream processing	41
edge effect	check for an edge effect	15
efficacy endpoint	specify the efficacy endpoint	33
evaluation license	scope an evaluation license	68
evidence gap	identify an evidence gap	6
evidence package	assemble the evidence package	59
exclusive license	define exclusive-license scope	50
exclusivity scope	narrow the exclusivity scope	68
execution risk	manage execution risk	59
experimental unit	identify the experimental unit	32
exposure	characterize exposure	32
exposure-response relationship	examine the exposure-response relationship	32
external validation	obtain external validation	24
field of use	limit the field of use	50
field restriction	define a field restriction	68
financing round	prepare for a financing round	60

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

fit for purpose	demonstrate fit-for-purpose performance	23
foreground IP	address foreground IP	50
freedom to operate	obtain a scoped freedom-to-operate review	51
functional readout	confirm a functional readout	6
generalizability	assess generalizability	6
go/no-go decision	support a go/no-go decision	59
good laboratory practice	verify good laboratory practice status	33
governance	define collaboration governance	69
governing agreement	identify the governing agreement	51
gross cash burn	distinguish gross cash burn	60
hold time	justify hold times	42
inflection point	define the proposed inflection point	60
intellectual property	identify relevant intellectual property	50
intermediate precision	evaluate intermediate precision	14
inventorship	assess inventorship	50
joint steering committee	establish a joint steering committee	69
know-how	identify transferred know-how	51
license	negotiate a license	50
license grant	delimit the license grant	68
limit of detection	establish a limit of detection	23
limit of quantification	establish a limit of quantification	23
lot-to-lot variation	evaluate lot-to-lot variation	15
mass transfer	assess mass-transfer behavior	42
material transfer agreement	finalize a material transfer agreement	51

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

matrix effect assess matrix effects	23
mean signal compare mean signals	14
mechanism of action investigate the mechanism of action	5
milestone consideration clarify milestone consideration	69
milestone criterion specify the milestone criterion	59
milestone payment verify milestone-payment conditions	60
mixing characterize mixing	42
modality compare therapeutic modalities	5
model relevance assess model relevance	32
negative predictive value interpret negative predictive value	24
net cash burn calculate monthly net cash burn	59
NOAEL interpret the NOAEL in context	33
nondisclosure agreement review a nondisclosure agreement	51
nonexclusive license discuss a nonexclusive license	50
normalization document normalization	15
off-target effect evaluate off-target effects	6
onward transfer address onward transfer	51
option agreement negotiate an option agreement	68
option exercise define option-exercise requirements	68
orthogonal assay run an orthogonal assay	5
ownership clarify ownership	50
partnering thesis articulate the partnering thesis	68
patent claim review relevant patent claims	51
patentability assess patentability	51

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

patient stratification	justify patient stratification	24
permitted use	define permitted use	51
pharmacodynamic biomarker	measure a pharmacodynamic biomarker	24
pharmacodynamics	evaluate pharmacodynamics	32
pharmacokinetics	characterize pharmacokinetics	32
plate effect	assess a plate effect	15
platform technology	develop a platform technology	5
positive predictive value	interpret positive predictive value	24
potency	evaluate product potency	41
precision	assess analytical precision	14
preclinical package	assemble a preclinical package	32
predictive biomarker	evaluate a predictive biomarker	24
prevalence	account for prevalence	24
process characterization	plan process characterization	42
process parameter	document process parameters	42
process validation	plan process validation	42
prognostic biomarker	distinguish a prognostic biomarker	24
proof of concept	define the proof-of-concept scope	6
proof of mechanism	seek proof of mechanism	6
prospective validation	design prospective validation	24
publication embargo	define a publication embargo	51
publication review	specify publication review	51
purity	assess product purity	41
quality-control sample	review quality-control samples	15

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

randomization	document randomization	32
readout	schedule a data readout	59
reagent lot	compare reagent lots	15
recovery	calculate downstream recovery	41
reference standard	qualify a reference standard	15
repeatability	assess repeatability	14
reportable range	specify the reportable range	23
reporter signal	quantify a reporter signal	5
reproducibility	define reproducibility conditions	14
reserved rights	identify reserved rights	68
resource allocation	justify resource allocation	59
rights reversion	define rights reversion	69
root-cause investigation	conduct a root-cause investigation	15
royalty	define the royalty basis	69
run metadata	preserve run metadata	15
run rate	qualify the run-rate estimate	60
safety margin	justify the safety-margin comparison	33
sample size	justify the sample size	32
scale-down model	qualify a scale-down model	42
scale-up	evaluate scale-up risks	42
scientific risk	reduce scientific risk	59
scientific thesis	articulate the scientific thesis	5
selectivity	assess selectivity	6
sex as a biological variable	address sex as a biological variable	32

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

standard deviation	calculate the standard deviation	14
strategic fit	assess strategic fit	68
study report	review the full study report	33
sublicensing	address sublicensing permission	68
target	define the target	5
target engagement	assess target engagement	5
technical replicate	distinguish technical replicates	5
technology transfer	execute technology transfer	41
term sheet	prepare a reviewed term sheet	69
termination right	review termination rights	69
territorial restriction	clarify territorial restrictions	68
territory	specify the territory	50
third-party rights	assess third-party rights	51
titer	report harvest titer	41
tolerability	report tolerability findings	33
toxicology	review toxicology data	33
translational hypothesis	test a translational hypothesis	6
translational uncertainty	communicate translational uncertainty	33
uncertainty interval	report the uncertainty interval	33
unrestricted cash	report unrestricted cash	59
upfront payment	distinguish an upfront payment	69
upstream processing	characterize upstream processing	41
within-run variation	quantify within-run variation	14
yield	define the yield basis	41

Original cases. Clear boundaries.

These are original fictional teaching conversations, not transcripts of real workers. Names, figures, organizations, and incidents are invented. Sources support terminology and communication aims, not the fictional results or an endorsement of this book.

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